

The Impact of Mortgage Rates on the Housing Market

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Abstract

We analyze the relationship between mortgage rates and housing activity in the US, focusing on house prices. Employing impulse response functions (IRFs) and forecast variance ratios (FVRs), we address potential endogeneity concerns by instrumenting mortgage rates with changes in long-term yields. Our findings reveal a significant impact of mortgage rate changes on house prices, with the maximal implied long-run semi-elasticity being close to 10. The FVRs provide set-identified variance decomposition for the instrumented shock, suggesting that mortgage rate shocks explain 2–25% of house price dynamics. The results also highlight substantial uncertainty, likely stemming from potential nonlinearities and regional factors influencing house price formation, which we further investigate. Additionally, we show that changes in mortgage rates significantly affect various dimensions of housing activity, including houses sold, construction, and building permits.

JEL: E32, E43, E52, R21, R31

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1 Introduction

The sensitivity of housing demand to mortgage rates is essential for understanding house price fluctuations. Since the 2007–09 financial crisis, the relationship between interest rates and house prices has garnered significant research attention. However, most macroeconomic studies exploring house price dynamics focus primarily on changes in monetary policy rates rather than mortgage rates. This focus introduces inherent limitations, as changes in policy rates do not directly translate to changes in mortgage rates; i.e., the link between short-term monetary policy rates and mortgage rates is generally weak (van Binsbergen & Grotteria, 2024). Even studies employing external instruments often emphasize surprises in short-term interest rates to identify monetary policy shocks, leaving gaps in understanding the direct effects of mortgage rates.

Contrary to the prevailing emphasis on policy rates, economic theory suggests that mortgage rates exert a more direct influence on housing demand and house prices. To address this gap, our study explicitly investigates the effects of mortgage rates on housing activity in the United States using an instrumental variable approach. By analyzing data spanning 1988 to 2023, we encompass a complete housing market cycle, providing insights into both boom and bust periods.

Our primary objective is to study the impact of changes in mortgage rates on housing variables, with a particular emphasis on house prices. First, we aim to quantify the effect of mortgage rates by employing impulse response functions (IRFs) and forecast variance ratios (FVRs) to assess the temporal effects and significance of mortgage rate changes on housing activity. To ensure the robustness of our findings, we address potential endogeneity issues by instrumenting mortgage rates using changes in long-term yields and applying advanced econometric methodologies, including instrumental variable local projections (LP-IV), internal Bayesian vector autoregression (internal VAR), proxy structural vector autoregressions (VAR-IV), and structural vector moving average models with external instruments (SVMA IV). These techniques reveal that mortgage rate shocks have a significant long-run effect on house prices, with estimated semi-elasticities reaching up to 10^1 or even more in some specifications, and that such shocks may explain up to 25% of house price dynamics. However, these effects typically manifest with substantial delays and are subject to considerable uncertainty.

Recognizing the heterogeneity of housing markets across the United States, we also examine regional variations in house price responses to mortgage rate changes. This part of our analysis allows us to identify potential nonlinearities and understand how regional characteristics shape these responses. In particular, we explore how housing supply elasticities² and regional interest rate levels mediate the effects of mortgage rate fluctuations. In more detail, our investigation focuses on whether house prices in regions characterized by differing average interest rate levels

¹Therefore, the results suggest that one-percentage point shock to mortgage rates could lead to a cumulative 10% change in house prices.

²In this part, our bears a resemblance to Aastveit & Anundsen (2022), and we generally confirm their findings. However, whereas their research predominantly investigates monetary shocks, our analysis focuses on the effects of changes in mortgage rates.

for new mortgages exhibit distinct sensitivities to changes in mortgage rates. Additionally, using supply elasticity estimates from Saiz (2010) and alternative instruments from Guren *et al.* (2021), we evaluate whether regions with tighter supply constraints show distinct price dynamics in response to mortgage rate changes. By doing so, we capture the interaction between housing supply constraints, regional interest rate conditions, and mortgage rate adjustments.

Finally, our investigation extends beyond house prices to encompass broader housing market activity. We analyze how mortgage rates influence housing sales, construction, and building permits. This comprehensive perspective enhances our understanding of the overall impact of mortgage rates on the housing market.

In summary, our study contributes to the literature by rigorously evaluating the impact of mortgage rates on house prices and broader housing activity through the use of an external instrument approach and multiple state-of-the-art methodologies. By differentiating between mortgage rates and monetary policy rates, we offer enhanced clarity in understanding housing dynamics and mortgage market behaviors. Additionally, we provide a comprehensive review of both empirical and theoretical findings regarding the effect of mortgage rates (or interest rates more broadly) on house prices, which serve as the primary variable of interest in this article.

The remainder of this paper is organized as follows: Section 2 reviews the existing literature. Section 3 presents theoretical models that provide intuition for house price semi-elasticities. Sections 4 and 5 describe the empirical methodology and the data, respectively. Section 6 discusses empirical results for house prices, while Section 7 extends the analysis to other housing variables.

2 Literature review

In this section, we provide a literature overview of the relationship between interest rates and house prices (as other housing variables receive significantly less scholarly attention).

Our primary focus lies in examining the effect of mortgage rates on house prices, with a clear distinction between mortgage rates and monetary policy rates. However, the distinction between these rates does not negate the impact of policy rate changes on house prices; on the contrary, monetary policy exerts influence through several channels, including but not limited to its effect on mortgage rates. Beyond short-term rates, monetary policy shapes long-term yields through mechanisms such as credit channels, expectations, and risk premium adjustments, which in turn impact consumption and investment decisions (Kuchler *et al.*, 2023). These channels operate with varying intensities and lags, complicating the task of isolating specific effects. For a comprehensive discussion of these channels and their influence on house prices, see WEO (2024).

Theoretical models often attempt to differentiate between the various channels, whereas empirical studies usually aggregate them due to practical constraints. Empirical research, particularly macroeconomic studies, tends to focus on (short-term) monetary policy shocks, as exogenous variation in mortgage rates independent of confounding factors is scarce. This aggregation can lead to ambiguity, with changes in policy rates frequently misinterpreted as direct

changes in mortgage rates, neglecting the complexities of rate pass-through mechanisms. For example, the trajectory of long-term yields reflects not only current and expected policy rates but also evolving trends in inflation, real interest rates, and term premiums, as highlighted by Bauer & Rudebusch (2020).³ Consequently, empirical estimates of interest rate effects on house prices often vary, with reported semi-elasticities reflecting aggregate impacts rather than the 'pure interest rate channel' representing the influence of mortgage rates on house prices.

This study contributes to the extensive literature on the transmission of interest rate changes to housing markets and the broader interaction between credit conditions and house prices. Seminal contributions in this field include works by Dokko *et al.* (2011), Mian & Sufi (2011), Favara & Imbs (2015), Jordà *et al.* (2015), Di Maggio *et al.* (2017), Sutton *et al.* (2017), and Bhutta & Ringo (2021), which collectively emphasize the roles of debt financing, credit supply, and credit standards in shaping the housing market dynamics. Moreover, studies by Berger *et al.* (2021), Aastveit & Anundsen (2022), Eichenbaum *et al.* (2022), Aastveit *et al.* (2023), and Batzer *et al.* (2024) focus on nonlinearities such as state dependence on interest rates due to mortgage rate lock-in effects⁴, the prevalence of prepayable mortgages, the interplay between adjustable and fixed-rate mortgages, housing supply elasticity, and other influencing factors.

Despite the breadth of the existing literature, the findings are highly heterogeneous. Many studies suggest that house price sensitivity to monetary policy changes is relatively modest. For example, Kuttner (2012) reports long-run semi-elasticities ranging from 3 to 8, with subsequent (meta-analysis) research often finding even lower estimates. Similarly, Williams *et al.* (2015) synthesize 11 studies, finding an average house price increase of 7% following a one-percentage point expansionary monetary policy shock, though estimates range from 1.7% to 10.8%. A meta-analysis by Ehrenbergerova *et al.* (2023), accounting for publication bias and model misspecifications, concludes that the maximal long-run semi-elasticity is below one, indicating that house price responses to policy rate changes are minimal. However, these findings often fail to directly address the semi-elasticity of house price changes to mortgage rates due to the imperfect transmission of policy rates.

However, even studies that explicitly account for mortgage rates tend to report low semi-elasticities. For instance, Fuster & Zafar (2021), using survey data, estimated the effect of the mortgage rate on willingness to pay, which corresponds to a semi-elasticity of approximately 2.5. Similarly, Adelino *et al.* (2012) used a difference-in-differences approach with a conforming loan limit in the US, which restricts the size of mortgages that Fannie Mae and Freddie Mac can purchase, to estimate semi-elasticities. They found them ranging from 1 to 9. The difference-in-differences approach was also employed by Davis *et al.* (2020) to analyze the impact of the Federal Housing Administration premium cut in January 2015; the authors estimate semi-elasticity to be 3.4. DeFusco & Paciorek (2017) identify the effect of interest rates on borrower behavior by exploiting exogenous variation in the conforming loan limits and estimate reduction

³For instance, reductions in the term premium or natural interest rate may significantly lower long-term mortgage rates even when short-term rates are near the zero lower bound.

⁴Mortgage rate lock-in effects lead to reduced home sales. The resulting supply reduction increases home prices, potentially outweighing the direct impact of elevated rates.

in total mortgage debt in response to a 1% rise in the first mortgage interest rate between 1.5 and 2%, which using the ability to pay approach roughly translate to house price semi-elasticity.

Table 1 summarizes the estimated effects of interest rates on house prices from key studies (building upon the work of Williams *et al.* (2015), Liu *et al.* (2021), and Ehrenbergerova *et al.* (2023)). For comparability, all results are rescaled to represent the impact of a one-percentage point increase in interest rates on house prices. The table offers the following insight: earlier research generally finds a modest relationship between interest rates and house prices, whereas more recent studies suggest stronger effects. This shift may reflect the adoption of more advanced econometric techniques and the availability of longer time series that include periods of heightened housing market volatility, such as the global financial crisis. It is also important to note that real house price dynamics were relatively subdued during the 1980s and 1990s, which may have contributed to the smaller estimates in earlier work.

Table 1: Estimated Effects of Interest Rates on House Prices in the Literature

Study	Specification	Maximal Effect	Initial Effect	Country	Interest rate	Period
Macro						
Aastveit & Anundsen (2022)	1 (expansion)	8.2	1.5	US	mortgage rate	1983-2007
Aastveit & Anundsen (2022)	2 (contraction)	0.5	0	US	mortgage rate	1983-2007
Aastveit & Anundsen (2022)	3 (expansion)	8.1	-0.2	US	short-term policy rate	1986-2008
Aastveit & Anundsen (2022)	4 (contraction)	6.3	0.1	US	short-term policy rate	1986-2008
Albuquerque <i>et al.</i> (2020)	1	2.2	0	US	short-term policy rate	1991-2019
Albuquerque <i>et al.</i> (2024)	1	16	0	US	short-term policy rate	1999-2019
Assenmacher-Wesche & Gerlach (2008)	1	2.4	0	US	short-term policy rate	1986-2006
Assenmacher-Wesche & Gerlach (2010)	1 (normal)	1.6	0	International	short-term policy rate	1986-2008
Assenmacher-Wesche & Gerlach (2010)	2 (boom)	2.2	0	International	short-term policy rate	1986-2008
Bauer & Granziera (2017)	1	3	1	International	short-term policy rate	1975-2014
Ben-David <i>et al.</i> (2019)	1	9	5	International	mortgage rate	1973-2018
Calza <i>et al.</i> (2013)	1	3.3	0	International	short-term policy rate	1980-2008
Cooper <i>et al.</i> (2022)	1	2.5	NA	US	short-term policy rate	2000-2007
Cooper <i>et al.</i> (2022)	2	6.6	NA	US	short-term policy rate	1980-2007
Chudik & Kumar (2023)	1	4.5	0.1	US	mortgage rate	1975-2023
Chudik & Kumar (2023)	2	2	0.1	US	mortgage rate	1975-1998
Chudik & Kumar (2023)	3	11	0	US	mortgage rate	1999-2023
Del Negro & Otrok (2007)	1	4	4	US	short-term policy rate	1986-2005
Demary (2010)	1	1.5	0.7	US	short-term policy rate	1980-2008
Dias & Duarte (2019)	1	0.5	0.1	US	short-term policy rate	1981-2017
Fischer <i>et al.</i> (2021)	1	14	NA	US	short-term policy rate	1997-2012
Gorea <i>et al.</i> (2023)	1 - 3	3	1	US	mortgage rate	2000-2019
Gupta <i>et al.</i> (2012)	1	1	0	US	short-term policy rate	1986-2003
Larson (2022)	1 (lower IR level)	10	0	US	mortgage rate	1996-2021
Larson (2022)	2 (higher IR level)	15	0	US	mortgage rate	1996-2021
Liu <i>et al.</i> (2021)	1	2	1	US	mortgage rate	1975-2020
Liu <i>et al.</i> (2021)	2	2.5	1	US	mortgage rate	1975-2020
McDonald & Stokes (2013)	1	1.6	0	US	short-term policy rate	1987-2010
McDonald & Stokes (2013)	2	0	0	US	mortgage rate	1987-2010
Musso <i>et al.</i> (2011)	1	2.5	0.6	US	short-term policy rate	1986-2009
Paul (2020)	1	5	0	US	short-term policy rate	1988-2017
Paul (2020)	2 (time-varying)	5	0	US	short-term policy rate	1988-2017
Sa <i>et al.</i> (2011)	1	1.2	-0.1	International	short-term policy rate	1984-2007
Smets & Jarociński (2008)	1	2	0	US	short-term policy rate	1987-2007
Smets & Jarociński (2008)	2	0-6	0	US	short-term policy rate	1987-2007
Vargas-Silva (2008)	1	2.4	0	US	short-term policy rate	1965-2005
Vargas-Silva (2008)	2	2.4	0	US	short-term policy rate	1965-2005
Williams <i>et al.</i> (2015)	1	6.3	0	International	short-term policy rate	1870-2013
Williams <i>et al.</i> (2015)	2	8.2	0	International	short-term policy rate	1946-2013
Micro						
Adelino <i>et al.</i> (2012)	1	1.3 - 5.3	NA	US	mortgage rate	1998-2008
Davis <i>et al.</i> (2020)	1	3.4	NA	US	mortgage rate	2015
DeFusco & Paciorek (2017)	1	1.5 - 2	NA	US	mortgage rate	1997-2007
Fuster & Zafar (2021)	1	2.5	NA	US	mortgage rate	2014

Note: The table shows the estimated effects of a 1% increase in interest rates on house prices.

Theoretical models, in contrast, often predict substantial impacts of interest rates on house prices. For example, the static user cost model of Poterba (1984) and Himmelberg *et al.* (2005) implies semi-elasticities exceeding 10 under most calibrations. Glaeser *et al.* (2012) demonstrate that when the above-mentioned approach is extended to include refinancing, the connection between the private discount rate and market interest rates, and volatile interest rates, it may substantially reduce the implied effect. (The baseline specification implies semi-elasticity between 8 and 16.) Additionally, a model by Garriga *et al.* (2019) illustrates that shocks to mortgage rates and leverage can account for significant changes in house prices, with potentially very large implied semi-elasticities. Nonetheless, the magnitudes depend on the duration of changes in financial condition. A simple alternative model based on the borrowing capacity of households (ability to pay) is presented by McQuinn & O'Reilly (2008) and others again indicate relatively large semi-elasticities. On the other hand, Anenberg & Kung (2017) introduce mortgages into a dynamic general equilibrium directed search model of the housing market, indicating relatively modest elasticities of house prices due to market frictions, although home sales and construction are much more elastic than house prices.

The apparent dichotomy between theoretical predictions and empirical estimates underscores the complexity of this relationship. While theoretical models often adjust assumptions to align with empirical findings, our results, as presented below, suggest that basic theoretical frameworks may provide reasonable approximations.

3 Theory

This section briefly reviews several theoretical models to deepen our understanding of the semi-elasticity of house prices to mortgage rates implied by these frameworks. We begin with the user cost approach, a common benchmark in theoretical housing market models. Next, we examine the borrowing-capacity (ability-to-pay) approach proposed by McQuinn & O'Reilly (2008), which emphasizes the role of credit constraints in determining house prices. Finally, we extend a partial equilibrium model (a mortgage-financed house purchase) from Garriga *et al.* (2017) to longer time horizons. The primary focus is on deriving and comparing the semi-elasticity of house prices to mortgage rates as predicted by these models, providing a theoretical foundation for interpreting our empirical findings.

Following the exposition by Kuttner (2012), the user cost approach is grounded in the no-arbitrage condition, positing that the cost of renting should equal the risk-adjusted cost of homeownership. In other words, the model considers the costs associated with owning a home, including mortgage interest, property taxes, maintenance, and expected capital gains or losses. This relationship can be written as follows:

$$\frac{R_t}{P_t} = [(i_t + \tau_t^p)(1 - \tau_t^y) + \sigma_t + \delta_t - \pi_t^e] - (g_t^e - \pi_t^e) \quad (1)$$

where $\frac{R_t}{P_t}$ denotes the rent-to-price ratio, π_t^e denotes expected inflation, i_t represents the nominal

mortgage rate⁵, δ_t is the depreciation rate, σ_t is the risk premium to compensate homeowners for the higher risk of owning versus renting, g_t^e is expected nominal house price growth, and τ_t^p , τ_t^y denote the property and income tax rates (to account for the deductibility of mortgage payments). The term in square brackets is referred to as the real user cost (UC).

Differentiating with respect to the mortgage rate and assuming no expected real house price growth yields semi-elasticity:

$$\frac{1}{P} \frac{\partial P}{\partial i} = -\frac{1 - \tau^y}{UC} \quad (2)$$

Most calibrations assume expected inflation between 2% and 3%, mortgage rate in the 4% to 7% range, τ_t^y between 20% and 25%, δ_t between 1% and 2%, and τ_t^p in 1% to 2.5% range. This implies real user cost (UC) is between 4% and 6%, ignoring the risk premium. Therefore, the implied semi-elasticity (1% shock to mortgage rates) is roughly between 13% and 20%.

McQuinn & O'Reilly (2008), Damen *et al.* (2016), Andrieu & Plašil (2019a,b), Biljanovska *et al.* (2023), and van der Drift *et al.* (2023) present a simple alternative approach to the user cost model called a borrowing-capacity (ability to pay) approach. The approach is based on how much housing a household can afford given its nominal income (Y_t), mortgage rate (i_t), and loan-to-value requirements (LTV_t).⁶ The implicit assumption is that the house price is determined by the marginal home buyer, characterized as a household that finances the house purchase with a mortgage but has otherwise limited access to financial markets (hand-to-mouth household).⁷ More specifically, a household allocates a portion θ_t of its income to service its mortgage payment ($m(i_t, M)$), which in turn determines the size of the mortgage loan (initial maturity of the mortgage loan is M years and we assume fixed rate annuity mortgage). The size of the mortgage loan plus the down payment determines the amount of attainable housing (house price), which can be rewritten using the LTV ratio as follows:

$$P_t = \theta_t Y_t m(i_t, M) + D_t = \frac{1}{LTV_t} \theta_t Y_t m(i_t, M) = \frac{1}{LTV_t} \theta_t Y_t \frac{(i_t + 1)^M - 1}{i_t (i_t + 1)^M} \quad (3)$$

The corresponding semi-elasticity is:

$$\frac{1}{P} \frac{\partial P}{\partial i} = -\frac{(i + 1)^{M+1} + (-M - 1) i - 1}{i (i + 1) \left((i + 1)^M - 1 \right)} \quad (4)$$

Assuming mortgage rates of 4%-7% and maturities of 20-30 years, this model suggests semi-elasticities close to 10%.

Our extended model, inspired by Garriga *et al.* (2017), considers a representative hand-to-mouth household over 31 periods receiving steady income Y . At time $t = 0$, the household

⁵We assume that foregone interest earned by investing in alternative asset is equal to the mortgage rate (accounting for tax deduction effect) and no down payment. For more nuanced view, see Glaeser *et al.* (2012) and Miles & Monro (2021).

⁶This can be equivalently expressed using the down payment amount (D_t) due to one to one mapping between LTV_t and D_t for maximum values of LTV_t and θ_t .

⁷Kaplan *et al.* (2014) and Cloyne *et al.* (2020) show that the typical homeowner is financially constrained, often exhibiting hand-to-mouth behavior.

makes a one-time house purchase. The implicit assumption is again that the house price is determined by the marginal home buyer, who finances the house purchase with a mortgage. The purchase is financed through a combination of mortgage debt and savings. The household allocates a fraction θ of the house price to mortgage debt (l), with a corresponding mortgage payment m , while the remaining fraction $1 - \theta$ is covered by their income in the first period as a down payment. The house price equals $l + D$. The mortgage can be used only for house purchases, and both the house and mortgage lasts for 30 periods.

The lifetime utility function is defined as follows: $U = \sum_{t=0}^{31} \beta^t \xi \log(c_t) + \sum_{t=1}^{31} \beta^t (1-\xi) \log(h)$, where β denote discount factor, c_t is non-housing consumption, h is housing, and ξ is the weight on non-housing consumption.⁸ The utility is maximized with respect to the sequence of budget constraints. For the $t = 0$, the budget constraint has the following form: $Y_t = (1 - \theta)h + c_t$. For the periods $t = 1, 2, \dots, 30$, the budget constraint in every period has the following form: $Y_t = m + c_t$, where m denotes mortgage payment. We again assume an annuity mortgage, and the size of the mortgage can be calculated as follows: $m = l \frac{i(i+1)^M}{(i+1)^M - 1}$, where $M = 30$. The household maximizes utility for every c_t and h , subject to per-period budget constraints. Equivalently, the household maximise its utility by choosing m and D , which in turn implies c_t and h . The model was solved numerically to obtain the implied semi-elasticities. We assume β equal to 0.9883 as in Garriga *et al.* (2017) and ξ equal to 0.6.

The borrowing-capacity (ability-to-pay) approach emerges as a special case, or corner solution, of the extended partial equilibrium model. This scenario occurs when households are constrained by the Loan-to-Value (LTV) ratio, which dictates the down payment size, and the Debt-to-Income (DTI) ratio, which limits mortgage payments relative to income. In this context, households might wish to allocate more resources to housing but are restricted by credit constraints.

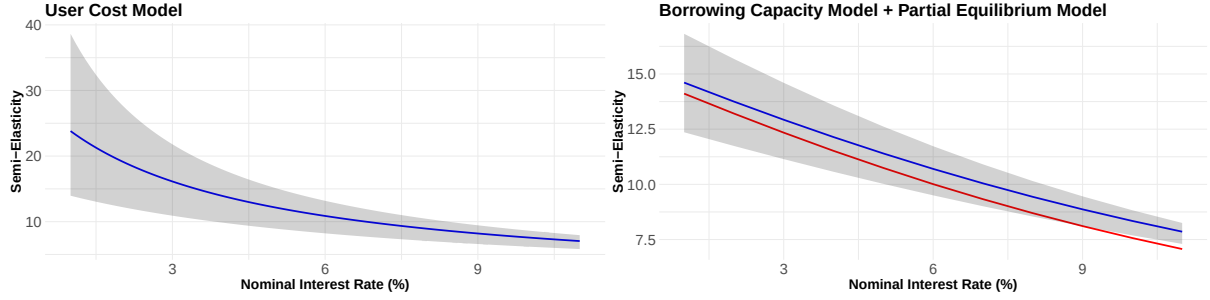
Figure 1 illustrates the implied semi-elasticities of house prices to changes in mortgage rates across different theoretical models. The left panel displays results from the user cost approach, where the blue line represents the baseline calibration, and the shaded area captures sensitivity checks. This model generally predicts a higher responsiveness of house prices to mortgage rate changes compared to the borrowing-capacity and partial equilibrium models shown in the right panel. In the right panel, the blue line corresponds to the borrowing-capacity model (equation 6, baseline calibration), with shaded areas reflecting sensitivity checks, while the red line shows the semi-elasticities from the partial equilibrium model. Both alternative models indicate somewhat smaller semi-elasticities that are less sensitive to mortgage rate fluctuations, with remarkably similar results between them. Sensitivity checks also reveal that the models, particularly the user cost model, are relatively sensitive to parameter changes. Additionally, Figure A1 in the Appendix provides complementary insights into the Loan-to-Value (LTV) and Debt-to-Income (DTI) ratios implied by the partial equilibrium model.

The theory also suggests two additional insights. First, the basic models outlined above

⁸This parameter mainly affects implied LTV and DTI ratios but has negligible impact on the interest rate semi-elasticity. The log utility function was chosen purely for convenience.

assume frictionless markets, where adjustments occur instantaneously in response to changes in economic conditions. Second, the semi-elasticity of house prices with respect to mortgage interest rates is non-linear and varies depending on the prevailing level of interest rates. These theoretical considerations should be taken into account in our empirical investigation, where we explore the role of market frictions and examine how the sensitivity of house prices evolves across different interest rate environments and over time.⁹

Figure 1: Theoretical Semi-Elasticities



Note: The left panel illustrates the user cost approach with a baseline calibration (blue line): $\tau_t^y = 25\%$, $\tau_t^p = 1.2\%$, $\sigma_t = 0$, $\delta_t = 1.5\%$, $\pi_t^e = 2.5\%$, $g_t^e = \pi_t^e$ and a sensitivity check (shaded area: $\tau_t^y = 15\% - 25\%$, $\tau_t^p = 1\% - 1.9\%$, $\sigma_t = 0 - 1\%$, $\delta_t = 1\% - 2\%$, $\pi_t^e = 2.5\% - 3.5\%$, $g_t^e = \pi_t^e\%$). The right panel presents the borrowing capacity approach under baseline calibration (blue line): $M = 30$, with sensitivity checks (shaded area $M = 25 - 35$) and the partial equilibrium model (red line, $M = 30$, $\beta = 0.9883$, $\xi = 0.6$).

4 Empirical Methodology

The relationship between interest rates and house prices, as well as broader housing activity, is frequently examined using empirical methods such as vector autoregressions (VARs), local projections, and panel data regressions across countries or regions. However, these approaches face challenges, particularly when interchanging mortgage rates with monetary policy rates and grappling with broader identification issues. To address these, we adopt an instrumental variable (IV) framework, relying on instruments that correlate with mortgage rate shocks but are otherwise uncorrelated with other shocks. Specifically, we use instrumental variable local projections (LP-IV), and internal or proxy structural vector autoregressions (VAR, respectively VAR-IV) to study these dynamics, measuring outcomes through impulse response functions (IRFs) and forecast variance ratios (FVRs). The former quantifies how a variable of interest (e.g., house prices or other housing activity variables) reacts over time to a one-time shock (e.g., mortgage rate changes)¹⁰, while the latter determines the proportion of variability in a variable explained by a specific shock. Specifically, we employ a structural vector moving average model with external instrumental variables (SVMA-IV) to estimate the (set-identified) forecast variance ratio (FVR).

⁹The impulse response functions (IRFs) estimated in the empirical analysis reflect the average semi-elasticity across varying levels of interest rates, as the prevailing mortgage rate has fluctuated over time. In other words, the estimated IRFs capture an average of the semi-elasticities across all periods in the sample.

¹⁰For instance, an IRF can show whether house prices immediately decline after a mortgage rate increase and how long this effect persists.

A critical component of our approach is the choice of a robust instrument to address endogeneity concerns, i.e., the availability of an instrumental variable that correlates with the mortgage rate shock but is otherwise uncorrelated with all other shocks. Mortgage rates both influence and are influenced by housing market dynamics, necessitating an instrument that isolates exogenous variation in mortgage rates. Institutional factors significantly affect the impact of interest rates on households, particularly the prevalence of fixed-rate versus adjustable-rate mortgages and liquidity constraints among borrowers. In the US, where long-term fixed-rate mortgages dominate the market, we use the 30-year fixed mortgage rate as the key variable of interest. This rate is instrumented using changes in the 10-year Treasury yield around Federal Open Market Committee (FOMC) announcements. This choice is motivated by the institutional structure of the US mortgage market and the observation that longer-term rates, such as the 10-year Treasury yield, play a more significant role than short-term policy rates in determining mortgage rates. For example, van Binsbergen & Grotteria (2024) have shown that following FOMC announcements, mortgage rates typically respond with a lag of about three weeks to changes in longer-term rates, moving almost one-for-one with 10-year swap rates.¹¹ These findings suggest that changes in longer-term rates are a strong instrument for studying the effects of mortgage rate changes.

Our approach builds upon the high-frequency identification of monetary policy surprises, a methodology pioneered by Kuttner (2001) and further developed in subsequent studies by Gürkaynak *et al.* (2005), Cieslak & Schrimpf (2019), Nakamura & Steinsson (2018a), Altavilla *et al.* (2019), Jarociński & Karadi (2020), Swanson (2021), Bauer & Swanson (2023a), and many others. We specifically utilize the data constructed by Swanson & Jayawickrema (2023) and used by Bauer & Swanson (2023b), which contains changes in asset prices during the narrow time window surrounding the FOMC announcements. The implicit assumption is that these movements primarily capture unexpected shifts in FED policy, including news about policy rates, asset purchases, and the general economic outlook. If markets are efficient, public information and systemic monetary policy responses are already incorporated into prices, ensuring that these movements reflect genuine monetary policy surprises rather than spurious variations.¹² By leveraging this high-frequency identification technique, we aim to isolate the causal impact of monetary policy surprises on mortgage rates and, subsequently, on house prices and housing activity in general.

While the use of changes in the bond yield around FOMC announcements as an instrument offers a robust approach to isolating exogenous variation in mortgage rates, it is not without potential limitations. A key challenge lies in ensuring that the instrument adequately captures all aspects of mortgage rate formation. The choice of the 10-year Treasury yield reflects its strong correlation with mortgage rates, yet it does not account for several other factors that influence mortgage rate dynamics, such as liquidity considerations, housing supply and demand

¹¹van Binsbergen & Grotteria (2024) illustrate that the correlation between 10-year government-bond par yield and 10-year swap rate is above 99%, and on FOMC days, the correlation is about 92%.

¹²This is not without a cost. The high-frequency shocks are usually relatively small, and the statistical power to assess the implied effects over multiple periods in the future is diminished due to the influence of other factors. For a comprehensive discussion, see: Nakamura & Steinsson (2018b).

shifts, or risk premiums. These omitted factors could introduce bias or weaken the causal inference if they are correlated with both the instrument and the housing market outcomes.¹³

One significant concern is the potential impact of monetary policy surprises on household income. For instance, changes in monetary policy may influence labor market conditions, and wage growth, which in turn affect housing demand and prices independently of mortgage rates. Similarly, stock prices and other asset valuations often respond to monetary policy shocks. To address these concerns, we demonstrate that omitted variables are not the main driving force for the observed results.¹⁴ We progressively add additional control variables to the regression model. By incorporating variables such as household income, unemployment rate, and asset price indices, we assess whether the estimated coefficients for the mortgage rate remain stable. Additionally, we also conduct robustness checks with alternative instruments to support the findings. Specifically, we also include 5-year and 30-year bond changes around FOMC announcements.

Another crucial consideration in our analysis is invertibility, which refers to the ability to recover structural shocks—the fundamental causes of changes in variables—from current and past values of the observed variables.¹⁵ If the shocks identified by our instruments cannot be adequately captured by the model, standard approaches like VAR-IV may yield biased results as they rely on the assumption that the shocks can be accurately recovered to study their effects. To address this, we use a methodology developed by Plagborg-Møller & Wolf (2022) to test for invertibility. A test for invertibility evaluates whether this assumption holds by checking if the mapping from shocks to observables can be inverted. Plagborg-Møller & Wolf (2022) showed that the external shock is invertible if and only if the instrumental variable does not Granger cause the vector of other series (pre-invertibility test). The authors also provide a continuous measure of the degree of invertibility, which is the R_0^2 value in a population regression of the shock on past and currently observed variables.¹⁶ If the shocks are not invertible, alternative methods (such as internal VAR and LP-IV) must be used to estimate the effects of interest.

To analyze the impact of mortgage rate changes on house prices and housing activity, we estimate impulse response functions (IRFs) using the local projection IV (LP-IV) method, an instrumental variable approach based on the two-stage least squares (2SLS) estimator. This methodology, introduced by Jordà (2005) and further refined by Jordà *et al.* (2015) and Stock & Watson (2018), directly captures the dynamic effects of shocks, making it possible to estimate the average effect of changes in mortgage rates on house prices and other housing variables. The observed shocks to Treasury yields around FOMC dates serve as instruments for mortgage rates, facilitating the estimation of their causal effects.

¹³In a similar fashion, Bauer & Swanson (2023b) address these concerns by orthogonalizing these surprises with respect to publicly available data preceding announcements. Their reassessment shows that high-frequency effects on asset prices remain robust.

¹⁴However, while our main findings remain directionally robust, we cannot fully disentangle the direct effects of interest rate changes from the indirect influence of other factors such as wage growth or stock price movements. This limitation warrants a degree of caution in the interpretation of our results.

¹⁵See, for example, Forni *et al.* (2019) and Chahrour & Jurado (2022) for more details.

¹⁶The shock is invertible if $R_0^2 = 1$, if $R_0^2 < 1$ the shock is non-invertible and VAR-IV gives biased IRFs. If shock is recoverable from all leads and lags of the endogenous variables, then $R_\infty^2 = 1$.

The method involves estimating a series of equations for different time horizons, i.e., for each horizon h ¹⁷, we estimate the following empirical specification:

$$Y_{t+h} = \alpha_h + \beta_h \Delta i_t + \gamma_h(L)W_{t-1} + trend_t + e_{t+h} \quad (5)$$

where Y_{t+h} represents the variable of interest at time $t+h$, such as the log of the Case-Shiller or FHFA house price index or another housing market variable. α_h and e_{t+h} denote intercept and idiosyncratic error term. Δi_t is the first difference of mortgage rate at time t instrumented by shocks to 10-year Treasury yields Z_t . The coefficients of interest, denoted by β_h , represent the direct impact of the change in mortgage rate for periods $h = 0, 1, \dots, 24$, corresponding to five years.

W_{t-1} and $\gamma_h(L)$ represent additional control variables and their estimated coefficients, consisting (in the baseline specification) of lagged values of unemployment (un_t) and inflation (π_t) and the lagged endogenous variable¹⁸, i.e., $W_t = (Y_t, \pi_t, un_t)$. For robustness checks, we also incorporate other variables as controls (wages, stock prices, housing activity). The inclusion of dependent variable lags is motivated by the findings of Herbst & Johannsen (2024) and Montiel Olea & Plagborg-Møller (2021), indicating that local projection can be biased and that lag-augmenting local projections enhances inference robustness and simplifies standard error calculations. $\gamma_h(L)$ is a polynomial in the lag operator of order 10 (regarding house prices). The lag structure is determined based on the Akaike Information Criterion (AIC). Specifically, for the (nominal) Case-Shiller Index, AIC indicates the use of 10 lags. When using the (nominal) FHFA index, the recommended lag order of 9. For real house price data, AIC recommends using 9 or 10 lags. For other variables that describe housing activity, AIC generally suggests around 8 lags, with the exception of houses sold, where 10 lags are used.¹⁹ Furthermore, to address potential nonstationarity, we incorporate a polynomial trend ($trend_t$), primarily linear, although sensitivity checks with quadratic trends yield consistent results, as per the guidance of Ramey (2016).

The LP-IV method's validity depends on the strength of the instrument.²⁰ To assess this, we compute the first stage effective (heteroskedasticity-robust) F-statistic from Olea & Pflueger (2013), as recommended by Andrews *et al.* (2019).²¹ While a common threshold for strong

¹⁷Instead of fitting a single complex model, LP-IV estimates the impact of a shock directly at each time horizon using simple regressions, one for each forecasted period. This approach avoids imposing potentially restrictive assumptions about the relationships between variables over time. To provide further intuition for the methodology, when the model is exactly identified, and there is one instrument (Z) and one treatment (Δi) (abstracting from control variables and with a slight abuse of notation), the LP-IV (2SLS) estimator can be expressed as a ratio of two sample covariances: $\beta_h = cov(Y_h, Z)/cov(\Delta i, Z)$. This formulation illustrates that the LP-IV estimator represents the ratio between the reduced-form and first-stage coefficients. Importantly, this equation holds for all horizons h (shifting Y by h periods), including the impact effect at $h = 0$. This offers a clear connection to the standard instrumental variables framework, as it directly aligns with the core IV logic of isolating exogenous variation through the use of a valid instrument (Inoue *et al.*, 2023).

¹⁸Given the inclusion of the lagged endogenous variable in the right-hand side, the IRFs should be interpreted as cumulative (Jordà, 2023).

¹⁹In robustness checks, we also vary the number of lags to ensure the stability of our results.

²⁰The validity of internal VAR and VAR-IV methods also depends on the strength of the instrument. For VAR models, the effective F-statistic corresponds to the first effective F-statistic period of LP-IV models.

²¹According to Andrews *et al.* (2019), the use of Kleibergen & Paap (2006) statistics is not recommended

instruments is an F-statistic above 10, Olea & Pflueger (2013) suggest a more stringent cutoff of 23. Below this threshold, they recommend using weak-instrument-robust confidence intervals. Given the potential for weak instruments, we employ the Anderson-Rubin (AR) statistic to construct confidence intervals. The AR statistic offers weak-instrument-robust inference, ensuring reliable confidence sets even when instrument strength is marginal. In particular, in our just identified model, AR-based inference does not sacrifice power compared to standard t tests under strong instruments (Andrews *et al.*, 2019). Confidence intervals are derived through point-wise inversion of the AR statistic, ensuring robust and reliable estimates of the impact of mortgage rate changes on the housing market. Furthermore, we also compute 2SLS standard errors using a heteroscedasticity-robust estimator.²²

Furthermore, we also provide several additional robustness checks regarding the LP-IV methodology. First, we simulate data using the estimated LP-IV model parameters and test whether our methodology can accurately recover the original parameters. Furthermore, following Young (2022), we calculate bootstrapped confidence intervals. This involves 50,000 replications of the following procedure: (1) sampling the data independently and with replacement from the original sample, and (2) computing the 2SLS coefficient for each replication. We specifically apply the bootstrap-t method; for more details, see Young (2022). Similarly, we also experiment with bootstrapping the errors on top of our simulated data to further assess the robustness of our approach and explore whether the results are similar to those obtained using the more standard bootstrap method. In other words, this residual bootstrap method consists of generating simulated data and then adding (resampled) measurement errors before re-estimating the model.²³

To further strengthen our result, we also calculate two alternative estimators. The first alternative estimator is the proxy structural vector autoregression (VAR-IV) used by Mertens & Ravn (2013), Gertler & Karadi (2015), Lunsford (2015), and Montiel Olea *et al.* (2021). This method leverages external instruments to identify structural shocks within an SVAR framework. Importantly, the instrument is not included directly in the reduced-form VAR. Instead, the structural shock is identified by projecting the instrument onto the reduced-form VAR residuals. More specifically, we estimate the following VAR(p) model: $X_t = \sum_{n=1}^p X_{t-n} + e_t$, where $X_t = (\Delta i_t, Y_t, \pi_t, un_t)$. The consistency of this estimator depends on the invertibility of the shock of interest; if the shock is not invertible, the estimator will not be consistent as illustrated in Forni *et al.* (2019) and Plagborg-Møller & Wolf (2021). We use ten or eight lags (depending on the variable of interest) for consistency with LP-IV, and we also include an exogenous linear trend. We estimate VAR-IV using the standard frequentist methodology. Given the possible weak instrument issue, we also display weak-instrument robust confidence

because there is no theoretical justification for their application in assessing instrument strength in non-homoskedastic settings. This limitation underscores the importance of relying on robust alternative statistics, such as the effective F-statistic proposed by Olea & Pflueger (2013), which is specifically designed to account for heteroskedasticity and provide more reliable guidance in evaluating instrument strength.

²²Newey-West (HAC) estimator with h lags provides similar results.

²³In this case, we simply calculate quantiles of our bootstrap 2SLS coefficients. (See, for example, Lütkepohl (2000) for a similar approach.)

intervals by Montiel Olea *et al.* (2021), which can be thought of as analogous to Anderson and Rubin’s (AR) confidence intervals for the LP-IV model. We also display confidence intervals calculated using the delta method as in Montiel Olea *et al.* (2021).

The second alternative estimator is an internal (recursive) Bayesian VAR (BVAR) that internalizes external instruments by including the instrumental variable in the reduced-form VAR estimation. Specifically, we estimate the following VAR(p) model: $(ZX)_t = \sum_{n=1}^p (ZX)_{t-n} + e_t$, where $(ZX)_t = (Z_t, \Delta i_t, Y_t, \pi_t, un_t)$. We specifically employ a Bayesian (B)VAR model with uninformative priors²⁴ (see Miranda-Agrippino & Ricco (2018)). We use 50 thousand draws to obtain credible intervals. Again, we include ten or eight lags and a linear trend. Plagborg-Møller & Wolf (2021) demonstrate that the local projection-IV (LP-IV) method is an invertibility-robust impulse response estimator and that an (infinite-lag) VAR, which includes the instrumental variable ordered first, yields the same impulse response as the LP-IV. This means that, unlike SVAR-IV, the internal (recursive) VAR provides consistent impulse response estimates even when the shock is non-invertible. However, if the shocks are invertible, VAR-IV offers substantially lower dispersion in its estimates compared to procedures that internalize external instruments, as evidenced by Li *et al.* (2024).

In addition to examining impulse response functions, we investigate the forecast variance ratio (FVR), which quantifies the reduction in forecast uncertainty achieved when the econometrician has full knowledge of the entire path of a specific shock—in this case, the mortgage rate shock (forecast error variance decomposition). To address potential issues of non-invertibility, we apply a structural vector moving average model with external instrumental variables (SVMA-IV) to estimate the FVR. This approach follows the methodology proposed by Plagborg-Møller & Wolf (2022), which allows for robust estimation without assuming invertibility.

The SVMA-IV estimator leverages a reduced-form, finite-order VAR model to approximate the second moments of the data. Specifically, this involves using the VAR-implied autocovariances and cross-covariances of the observed variables and the projection residuals. This approach eliminates the need for a structural VAR model, bypassing the assumption that the shocks can be fully captured by the current and lagged values of the observed variables. To derive the FVR, Plagborg-Møller & Wolf (2022) introduce a procedure that constructs identification bounds. These bounds are then incorporated into the confidence interval construction method developed by Imbens & Manski (2004), ensuring robustness to weak identification. The lower bound of the FVR is influenced by the strength of the instrumental variable, as a stronger instrument provides a more precise relationship between the shock and the observed variables, thereby increasing the lower bound. Conversely, the upper bound is determined by how informative the included variables are for capturing the shock’s dynamics. Under the additional assumption of invertibility, where the shock is fully spanned by the current and past values of the time series, variance decomposition becomes point-identified. This allows for more precise attribution of forecast variance to specific shocks. By combining these approaches, we can assess

²⁴The posterior estimates, therefore, are centered at the MLE/OLS.

the extent to which mortgage rate shocks drive house price dynamics, providing deeper insights into their role in the housing market.

To further investigate regional heterogeneity in the response of house prices to mortgage rate changes, we employ a panel data version of the instrumental variable local projection (LP-IV) approach. The empirical specification is as follows:

$$Y_{i,t+h} = \alpha_{i,h} + \beta_h \Delta i_t + \Theta_h(\Omega_i - \bar{\Omega}) \Delta i_t + \gamma_h(L) W_{i,t-1} + trend_t + e_{i,t+h} \quad (6)$$

where $Y_{i,t+h}$ denotes the log of house prices in the MSA i at time $t + h$. The term $\alpha_{i,h}$ represents unobserved MSA-specific effects, while $W_{i,t-1}$ represents additional control variables consisting of lagged values of unemployment ($un_{i,t}$) in the MSA, nationwide unemployment (un_t), nationwide inflation rate (π_t) and the lagged house price for the given MSA, i.e., $W_{i,t} = (Y_{i,t}, \pi_t, un_t, un_{i,t})$. The rest of the notation is the same as in the LP-IV case.

The main difference against LP-IV is the inclusion of the interaction term $(\Omega_i - \bar{\Omega}) \Delta i_t$, where Ω_i denotes the supply elasticity by Saiz (2010) or housing supply instrument by Guren *et al.* (2021). (The variables are expressed relative to the mean and rescaled to have unit variance. The interaction of change in long-term yields and supply elasticity instrument is used as another instrument.) This allows us to decompose the average effect of change in mortgage rates on house prices into direct effect (β_h) and indirect part related to the metropolitan area-specific supply elasticity (Θ_h), i.e. varying supply elasticity (indirect) effect. Consequently, we demonstrate that regions with more constrained housing supply exhibit a more pronounced reaction to mortgage rate shocks.

Alternatively, we also included the lagged value of regional interest rate levels for new mortgages as another interaction term $\Phi_{i,t-1}$. The modified empirical specification is given by:

$$Y_{i,t+h} = \alpha_{i,h} + \beta_h \Delta i_t + \Psi_h(\Phi_{i,t-1} - \bar{\Phi}) \Delta i_t + \gamma_h(L) W_{i,t-1} + trend_t + e_{i,t+h} \quad (7)$$

As before, the variable $\Phi_{i,t-1}$ is demeaned and standardized to have unit variance, also note that the interaction term (and regional interest rate levels) vary over time.²⁵ (The rest of the notation is the same as in the previous case.) This approach enables us to examine potential non-linearities in house price responses in relation to interest rate level, as suggested by theoretical models. The analysis reveals that areas with higher average interest rates exhibit a less pronounced response to changes in interest rates, highlighting the importance of regional interest rate levels in moderating the sensitivity of housing markets. Therefore, the interpretation is the same as in the housing supply case, i.e., we decompose the average effect of change in mortgage rates on house prices into direct effect (β_h) and indirect part (Ψ_h).

The approach was proposed by Cloyne *et al.* (2023), and it is an extension of Kitagawa-Blinder-Oaxaca decomposition into a local projection framework. Cloyne *et al.* (2023) further

²⁵The results were nearly identical when we employed a four/eight-quarter lagged moving average of regional interest rate levels, suggesting that regional factors exhibit persistence and do not experience abrupt changes. For this approach to be valid, it is essential that regional differences remain stable over time and are not significantly influenced by overall changes in mortgage rates, which appears to hold true based on our analysis.

note that the Ω_i needs to be exogenous, or the decomposition requires further identification assumptions. More generally, Gonçalves *et al.* (2024) showed that the state-dependent local projection recovers the population response regardless of the shock size only when the state is exogenous. Given the nature of the supply elasticity instruments, we believe that they can be considered quasi-exogenous.²⁶ Similarly, the regional interest rate levels are lagged to be quasi-exogenous to mortgage rate shock, and we explicitly control for regional unemployment levels. The model coefficients are estimated with a fixed effects estimator (FE) with oneway fixed effects. Standard errors are obtained using a two-way clustering at the metropolitan area and time level.²⁷

We acknowledge that using a panel local projection approach with lagged explanatory variables as controls introduces potential endogeneity bias. This bias, as discussed by Nickell (1981) and Mei *et al.* (2023), stems from the demeaning transformation applied to remove unobserved heterogeneity in the panel data framework. However, this bias tends to decrease considerably with an increasing number of observations. Simulations conducted by Judson & Owen (1999) suggest that the bias is minimal in panels with more than 30 observations. For this reason, we do not use GMM dynamic panel estimation.

5 Data

Our empirical analysis primarily leverages a dataset of quarterly time series covering the period from 1988:Q2 to 2023:Q4, allowing us to study house price dynamics over the last three decades. The primary variables include changes in mortgage interest rates, the logarithms of real and nominal house prices, and shocks to long-term yields. Control variables include the consumer price index (CPI), unemployment rates, and, in some specifications, wages and stock prices. Additionally, we analyze other housing activity indicators, such as housing permits, construction, and the number of houses sold or listed for sale.

For house price data, we rely on the Case-Shiller index, a weighted repeat sales index that tracks average price changes across arm's-length single-family home transactions. This index includes homes financed with nonconforming mortgages such as jumbo and subprime loans. To ensure robustness, we also use the FHFA (Federal Housing Finance Agency) house price index, another repeat sales index based on data from mortgages purchased or securitized by Fannie Mae or Freddie Mac. Unlike the Case-Shiller index, the FHFA index equal-weights house prices and includes refinances. The FHFA dataset is available from 1991:Q1, and we use it for robustness checks from 1991:Q1 to 2023:Q4. Both indices are seasonally adjusted, deflated using the CPI to obtain real house price indices, and log-transformed.

For a detailed analysis using panel data, we utilize the Freddie Mac House Price Index, which

²⁶Saiz (2010) instrument is an expression of the geographical and regulatory limitations that should be, therefore, exogenous to mortgage rates. The supply instrument by Guren *et al.* (2021), on the other hand, explicitly controls for changes in interest rates when calculating regional sensitivities.

²⁷We also calculated Driscoll-Kraay standard errors, which yielded similar results and were not reported further. Given the use of multiple instruments, we do not use Anderson-Rubin (AR) statistics to construct confidence intervals.

provides extensive house price information across different US regions. Leveraging MSA-level data enables us to gain granular insights into localized housing market dynamics. For robustness checks, we also employ the FHFA dataset (100 Largest Metropolitan Statistical Areas Purchase-Only dataset), which, unlike the Freddie Mac dataset, excludes appraisal values but features a smaller sample size. Log transformations and CPI adjustments are consistently applied to house price data in both datasets to ensure comparability, using seasonally adjusted end-of-quarter values.

House price data for metropolitan statistical areas (MSAs) are used to connect our findings with prior research, such as that by Saiz (2010) and Guren *et al.* (2021). Saiz (2010) developed an instrument based on geographic constraints, such as variations in land unavailability, that restrict housing supply. These elasticities are derived from topographical measures of undevelopable land and from regulatory supply restrictions captured by the Wharton Regulatory Land Use Index (WRLURI). WRLURI assesses MSA-level regulatory restrictions, such as complexities in obtaining building permits, while the topographical measure quantifies land availability constraints.²⁸ On the other hand, Guren *et al.* (2021) constructed an instrument reflecting the systematic sensitivity of local house prices to regional housing cycles. This is based on the premise that house prices in some cities are systematically more sensitive to regional house-price cycles than in others. This instrument is constructed by estimating the historical sensitivity of local house prices to regional housing cycles, capturing the systematic differences in how local markets respond to broader housing trends.²⁹ This additional analysis allows us to study if regions with more constrained housing supply experience a more pronounced response to mortgage rate shocks. Our (balanced) panel data covers 342 MSAs for the Guren *et al.* (2021) instrument, and 243 MSAs for the Saiz (2010) instrument, spanning 1993:Q1 to 2023:Q4, with 132 observations per unit.³⁰

The data regarding regional mortgage rates for new mortgages are obtained from the Single Family Loan-Level Freddie Mac Dataset. The dataset allows for the matching of metropolitan statistical areas or metropolitan divisions generally as of the loan funding date. This dataset includes loans securitized into Freddie Mac mortgage-backed securities (MBS) and provides loan-level origination data, enabling the calculation of interest rate levels at origination. We specifically use standard datasets, and we include only fixed-rate mortgages. The property type is either a condo or single-family (to be broadly comparable with house price data), and we do not include refinanced loans. To address outliers, the dataset is winsorized at the 99% level based on Loan-to-Value (LTV), Debt-to-Income (DTI) ratios, and interest rate at origination. The data are aggregated to a quarterly frequency³¹ at the MSA level, covering the period from 1999:Q1 to 2023:Q4, resulting in a balanced panel of 342 MSAs, respectively 81 MSAs if we

²⁸The data are publicly available online from the MIT Urban Economics Lab.

²⁹We multiply the Guren *et al.* (2021) instrument by minus one for it to be directly comparable with Saiz (2010) instrument. The data are publicly available on Professor Adam Guren’s website.

³⁰If we use the FHFA dataset, our panel data covers 81 MSAs for the Guren *et al.* (2021) instrument, and 74 MSAs for the Saiz (2010) instrument. The number of metropolitan statistical areas (MSAs) is lower than 100 because we were not able to match every metropolitan area with instruments by Saiz (2010) or Guren *et al.* (2021).

³¹We use simple unweighted means, though weighted means by loan size yield similar results.

use FHFA data. This analysis helps investigate whether regions with higher interest rate levels exhibit a more pronounced response to mortgage rate shocks.

Other data regarding housing activity include building permits (FRED code: PERMITNSA), family homes for sale (FRED code: HNFSEPUSSA), family houses sold (FRED code: HSN1FNNSA), and construction started (FRED code: HOUST). We aggregate data to a quarterly frequency by summing monthly data over the quarter. We use seasonally adjusted data whenever available. If seasonally adjusted data were not available, we adjusted them using the X-13ARIMA-SEATS method.

Unemployment rates and CPI data are sourced from the FRED database or directly from the Bureau of Labor Statistics. For nationwide analysis, we use seasonally adjusted end-of-quarter CPI values (FRED code: CPIAUCSL) to compute year-on-year inflation rates and end-of-quarter unemployment rates (FRED code: UNRATE). However, our findings were nearly identical when using (annualized) quarter-on-quarter inflation rates. In the panel specification, we also use MSA-specific seasonally adjusted end-of-quarter unemployment rates. Real wages are measured using seasonally adjusted end-of-quarter real 'Employed full-time: Median usual weekly real earnings: Wage and salary workers' (FRED code: LES1252881600Q). For stock prices, we rely on the (real) S&P 500 index, adjusted by CPI, sourced from Robert J. Shiller's publicly available dataset. We use end-of-quarter values in levels. These variables (and also other data related to housing activity) are included as controls to mitigate potential endogeneity concerns. For example, monetary policy changes can directly affect unemployment, wage growth, housing supply, and asset valuations. Monetary policy also influences inflation, which, in turn, impacts all nominal variables. These factors can independently affect housing demand and prices, aside from the direct effects of mortgage rates.

Mortgage rate data were sourced from the FRED database, and we specifically used the end-of-quarter 30-year mortgage rate (FRED code: MORTGAGE30US) given the prevalence of fixed-rate mortgages in the US (see WEO (2024)). The data are collected weekly through a survey by Freddie Mac. The survey captures mortgage rates for first-lien, conventional, conforming, and purchase mortgages, assuming a borrower with an 80% loan-to-value ratio and excellent credit. For our analysis, we utilized end-of-quarter values in levels.

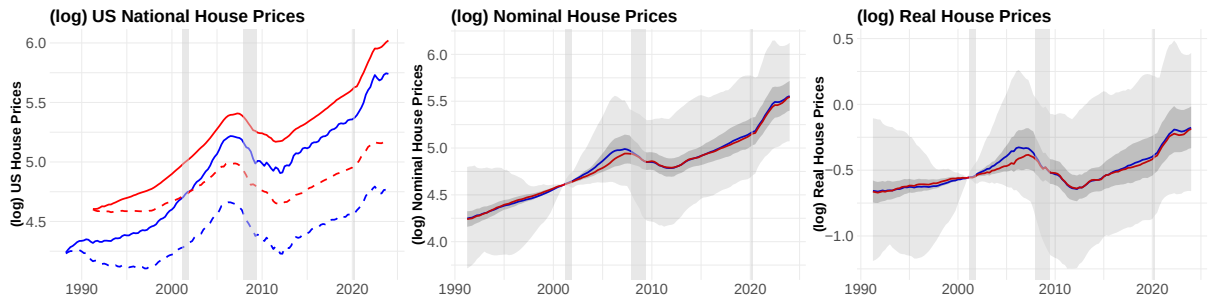
Finally, we utilized the (updated) high-frequency monetary policy surprises dataset by Swanson & Jayawickrema (2023), also used by Bauer & Swanson (2023b), to obtain shocks to long-term yields. This dataset comprises intraday changes in asset prices and yields, and we specifically use the intraday changes in 10-year Treasury yields, measured from Treasury note futures. The dataset covers both FOMC announcements and speeches by the Federal Reserve Chair. For comparability, we focus solely on FOMC announcements. To facilitate our analysis, we aggregate long-term yield surprises to a quarterly horizon. For a robustness check, we also use 5-year and 30-year Treasury yields intraday changes around FOMC dates.

6 Results - House Prices

In this section, we present our empirical findings regarding house prices. We start with an overview of general facts and descriptive statistics related to house prices, mortgage rates, and monetary shocks. Following this, we discuss the results derived from aggregate nationwide data. We then employ a panel data approach to confirm our findings and examine the effects of housing supply and differences in interest rate levels.

Figure 2 illustrates the evolution of house prices during the study period, highlighting both nationwide and regional patterns. The left panel shows the logarithm of US nationwide house prices in nominal and real terms (rescaled), while the middle and right panels display nominal and real house prices for MSAs, respectively.³² To enhance clarity, the figures include the mean, median, interquartile range, and minimum-maximum values. Nationwide real house prices exhibit a persistent trend, which aligns with the 'ability to pay' framework, suggesting that with stable mortgage rates and loan-to-value (LTV) ratios, real house prices should move proportionally with real household income, keeping the price-to-income ratio stable as household income reflects inflation and technological growth. At the MSA level, significant variability is evident, emphasizing the importance of regional dynamics in house price developments. These regional differences highlight the influence of local economic conditions, supply constraints, and housing market characteristics.

Figure 2: House Prices



Note: The left panel shows US nationwide house prices (red line: FHFA House Price Index, blue line: FHFA House Price Index, solid line: nominal index, dashed line: real index). The middle panel shows nominal MSA house prices (Freddie Mac dataset, red line: mean, blue line: median, dark shaded area: interquartile range, light shaded area: min-max range). The right panel shows real MSA house prices (Freddie Mac dataset, red line: mean, blue line: median, dark shaded area: interquartile range, light shaded area: min-max range). Shaded regions denote NBER-dated recessions.

For a more rigorous analysis of regional differences, Table 2 provides basic summary statistics (`xtsum` command) for the house price data (Freddie Mac dataset) used in the panel estimation. (Data regarding FHFA data are shown in Table A1.) The data include both nominal and real house prices, expressed as log first differences (multiplied by 100), with real house prices also analyzed in levels. The table reports key descriptive metrics: mean, standard deviation, minimum, and maximum values. These are further disaggregated into overall, between, and

³²We display results for 342 metropolitan areas, i.e., the Freddie Mac house price dataset obtained when combined with the instrument from Guren *et al.* (2021).

within components. A key observation is that the standard deviation between MSAs is similar to the standard deviation over time within districts. This observation underscores the substantial variability in the dataset, consistent with the broader descriptive patterns discussed earlier.

Table 2: Descriptive Statistics: House Prices (Freddie Mac Dataset)

Variable		Mean	Sd.	Min	Max	N
(log) Real House Prices	Overall	-0.499	0.21	-1.25	0.49	45144
(log) Real House Prices	Between		0.10	-0.79	-0.12	342
(log) Real House Prices	Within		0.18	-1.13	0.28	132
Δ (log) Nominal House Price	Overall	0.99	1.68	-15.21	12.81	44802
Δ (log) Nominal House Prices	Between		0.19	0.59	1.55	342
Δ (log) Nominal House Prices	Within		1.67	-15.12	12.83	131
Δ (log) Real House Prices	Overall	0.366	1.63	-16.16	11.85	44802
Δ (log) Real House Prices	Between		0.19	-0.04	0.92	342
Δ (log) Real House Prices	Within		1.62	-16.07	11.87	131

Note: The table displays summary statistics regarding the house price data used in the panel estimation. Sd. denotes standard deviation. N denotes the number of observations. Data regarding the Freddie Mac dataset for metropolitan statistical areas. The sample is 1993:Q1-2023:Q4.

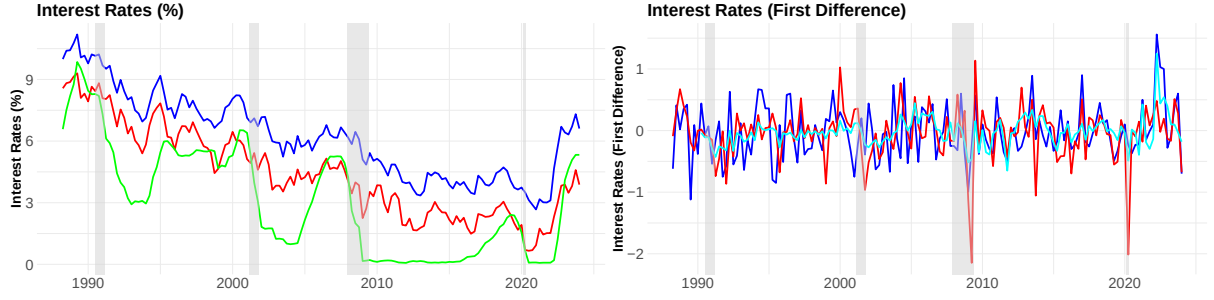
Figure 3 illustrates the development of interest rates, both in levels (left panel) and differences (right panel). The left panel shows that mortgage rates generally track the 10-year Treasury yield, maintaining a relatively stable spread above it, whereas their relationship with the Federal funds rate is less direct. In the right panel, we observe the changes in mortgage rates and their relationship with shocks to the 10-year Treasury yield (adjusted to match the variance of the mortgage rate). Additionally, in the figure, we also show a fitted mortgage rate³³, which tracks the fluctuations in the long-term yield shocks.

Table A2 in the Appendix provides first-stage regression results, highlighting the importance of using shocks to the 10-year Treasury yield around FOMC dates as an instrument. Comparisons with (short-term) monetary policy shocks, calculated following Nakamura & Steinsson (2018a), show that these shocks are significantly weaker instruments for mortgage rates. As a robustness check, the figure also includes results obtained using 5-year and 30-year Treasury yield changes around FOMC dates as alternative instruments. While these instruments exhibit similar explanatory power, the 10-year yield shock proves to be the most robust and is used in the primary specification. (The 30-year yield changes show somewhat higher explanatory power than the 5-year yield shock.) Nonetheless, results for the 5-year and 30-year yield shocks are also shown in another robustness check. Table A3 in the Appendix further demonstrates the robustness of the 10-year yield shock, as well as other variables as instruments, across subsample periods, both before and after the Great Financial Crisis.³⁴

³³First stage regression results for the following empirical specification: $Y_t = \alpha + \beta_h \Delta i_t + \gamma_h(L)W_{t-1} + trend_t + e_t$, where Y_t denote nominal Case-Shiller index.

³⁴Due to the shorter subsample periods, we restrict the number of lags for control variables to 4.

Figure 3: Interest Rates



Note: The left panel shows interest rates in levels (red line: 10-Year Treasury yield, blue line: 30-Year Fixed Rate Mortgage rate, green line: Federal Funds Effective Rate). The right panel shows interest rates in first differences (red line: 10-Year Treasury yield FOMC shock (rescaled), blue line: 30-Year Fixed Rate Mortgage rate, cyan line: Fitted 30-Year Fixed Rate Mortgage rate). Shaded regions denote NBER-dated recessions.

Figure 4 (left panel) presents the Impulse Response Functions (IRFs) of nominal house prices, as measured by the Case–Shiller index, in response to a one-percentage point increase in mortgage rates. The IRFs are derived using LP-IV, VAR-IV, and internal (Bayesian) VAR methodologies, which estimate the dynamic effects of mortgage rate shocks. Point estimates are shown as blue lines, with the 90% and 68% confidence (credible) intervals represented as gray bands. For the LP-IV and VAR-IV methods, these confidence intervals are weak-instrument robust. In some cases, dotted red lines provide alternative confidence bands. The analysis also includes first-stage effective F-statistics, calculated following Olea & Pflueger (2013), to assess instrument strength. Additionally, set-identified forecast variance ratios (FVRs), along with their 90% confidence bands, are displayed to evaluate the contribution of mortgage rates to house price dynamics.

The results indicate that a one percent shock to mortgage rates has a statistically significant effect on house prices (considering 90% HC confidence intervals or 68% weak-instrument confidence intervals). However, the effective F-statistics computed according to Olea & Pflueger (2013) indicate limited instrument strength, ranging between 4 and 9. These values fall below both the commonly accepted threshold of 10, and the robust threshold of 23 recommended by Olea & Pflueger (2013). Consequently, we emphasize weak-instrument robust confidence intervals in our analysis.

From an economic standpoint, the impact of mortgage rate shocks on house prices is substantial but delayed. Both LP-IV and internal (Bayesian) VAR models yield similar results. The short-run effect (within the first year) is approximately zero or even slightly positive but becomes significantly negative in the second year following the shock. The maximal semi-elasticity implied by the LP-IV and internal VAR models is approximately 8, with this negative effect dissipating by the fifth year. In contrast, the VAR-IV model suggests a significantly smaller effect, with a semi-elasticity of nearly half that size.

However, the Granger causality test (pre-invertibility test), as well as the continuous measure of the degree of invertibility R_0^2 , implies that the mortgage rate is not fully invertible, which means that VAR-IV may be severely biased. The p-value for the Granger causality test

is approximately 0.1 (F test statistic is around 15), and R_0^2 degree of invertibility is around 37% (upper estimate) as displayed in the table A4 in the Appendix. We can, therefore, reject invertibility (at the 10% level) since the confidence interval excludes 1.³⁵ This finding implies that the VAR-IV model is likely biased and underscores the importance of relying on invertibility-robust SVMA-IV analysis for variance decomposition. The FVR analysis suggests that mortgage rate shocks contribute minimally to house price dynamics in the first two years, but their contribution gradually increases over time, with the set-identified FVR indicating a range of 2% to 8% in later years (the upper bound of confidence intervals is approximately 25%, and the lower bound is zero). Therefore, we can rule out that the long-term yield shock is completely unimportant.³⁶

The right panel of Figure 4 extends the analysis to real house prices (Case-Shiller index), revealing results closely resembling those observed for nominal house prices. The key difference is that real house prices demonstrate a more rapid response, with impulse responses approximately 1–2 percentage points lower in the first two to three years. This suggests that real house prices react more quickly to changes in mortgage rates than nominal house prices. For the LP-IV model, the maximal semi-elasticity is approximately 6, while for the internal VAR model, it remains around 8. Conversely, the VAR-IV model estimates a lower semi-elasticity of roughly 3, likely reflecting bias from non-invertibility, as shown in Table A4 in the Appendix. The effective F-statistics are also somewhat weaker in the case of real house prices. However, the forecast variance ratio (FVR) suggests a slightly more significant role for mortgage rate shocks in influencing house prices, with contributions ranging from 3% to 10% (the upper bound of confidence intervals is approximately 25%, and the lower bound is zero).

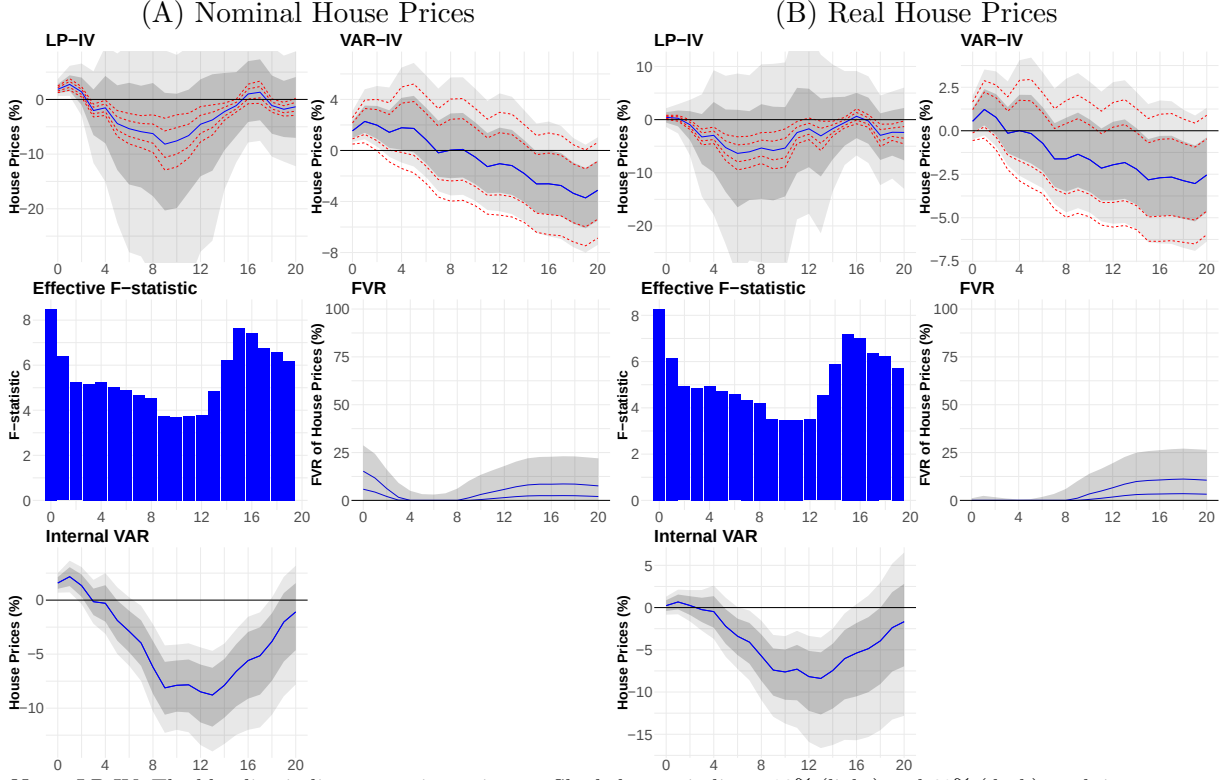
In summary, the results indicate that mortgage rates significantly influence house prices. However, the short-run effect is muted, likely because mortgage approval can take up to 60 days. Additionally, the Case-Shiller index employs a three-month moving average approach, meaning, for example, that January data represent the average for November-January. In the medium and long run, the effect is both significant and enduring. This outcome aligns with intuition from most theoretical models, although we are close to the lower bound of the estimates. However, it should also be noted that (Anderson-Rubin) weak-instrument robust confidence intervals, or even standard credible intervals, generally indicate very high uncertainty about the accuracy of the parameter estimate. For instance, LP-IV 90% weak-instrument robust confidence intervals are generally not statistically different from zero. Based on the results above, we suggest that the maximal implied semi-elasticity of house prices with respect to mortgage rates ranges typically between 5% and 15% (also taking into account robustness checks shown below). While standard 2SLS confidence intervals in the LP-IV case provide narrower intervals, we advise against using them due to the limited instrument strength. Furthermore, theoretical models generally suggest that the effects of mortgage rate changes are nonlinear. Our empirical impulse response functions (IRFs) reflect average semi-elasticities over varying interest rate levels, given

³⁵For all other specifications, we can also reject invertibility (see table A4).

³⁶The relatively wide range of possible results is due to the lower bound of the FVR being influenced by the strength of the instrumental variable, as a stronger instrument increases the lower bound. Conversely, the upper bound is determined by how informative the included variables are for capturing the shock's dynamics.

that mortgage rates have fluctuated significantly over the study period. Additionally, over the last thirty years—the period covered by this study—substantial changes in interest rate levels and significant institutional shifts have occurred. Additionally, the robustness of our instruments plays a critical role in shaping the results. These factors contribute to heightened uncertainty regarding the estimated effects, warranting caution in interpreting the results.

Figure 4: Impulse Response Functions (IRFs), Forecast Variance Ratio (FVR), and F-statistics for House Prices (Case–Shiller Index)



Note: LP-IV: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals constructed by inverting Anderson-Rubin (AR) tests. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the 2SLS estimator with HC standard errors. Effective F-statistic: First stage Effective F-statistic is computed as is Olea & Pflueger (2013) to assess instrument strength. Internal VAR: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) credible intervals. VAR-IV: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the delta-method. FVR: Forecast variance ratio, The blue lines indicate interval-identified estimates. Shaded areas indicate 90% confidence intervals. 10 lags were used in all models. The sample is 1988:Q2–2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

For robustness, we also used the FHFA nationwide house price index (Figure A2 in the Appendix). The results from the FHFA data largely confirm our findings with the Case–Shiller index. Notably, all three estimators (LP-IV, VAR-IV, and internal VAR) produce relatively similar results, though the VAR-IV estimator remains slightly lower than the other two. A key distinction emerges in the first year: using the FHFA data, house prices show a slight immediate decline after the shock, as opposed to being flat or slightly positive. In the nominal case, the LP-IV and internal VAR models estimate a maximal semi-elasticity of approximately 10, while the VAR-IV model estimates it at around 6. For the real FHFA index, the maximal

semi-elasticity is approximately 7 for the LP-IV and internal VAR models and about 5 for the VAR-IV model. Additionally, the FVR indicates that the contribution of mortgage rate shocks to nominal house prices ranges from 10% to 25%, while for real house prices, it ranges from 10% to 20%.

As an additional robustness check, we experimented with an expanded set of covariates, alternative lag structures, different instruments, and the inclusion of quadratic trends. Specifically, we extended the information set by incorporating variables that capture housing activity. The control variables included the house price index, unemployment rate, inflation, linear and quadratic trends, building permits, construction starts, family houses sold, and homes for sale. We also added real wages and stock valuations as supplementary covariates. Given a large number of estimated models, we utilized only the LP-IV method.³⁷ Figures A3 and A4 in the Appendix present results for the Case-Shiller and FHFA indices, respectively, demonstrating that the findings remain generally robust to these variations. While specific estimates differ, the maximal elasticity consistently falls within the range of 5% to 15%. The inclusion of different numbers of lags, quadratic trends, or housing activity variables has minimal impact on the results. However, the addition of real wages and stock valuations tends to accelerate the effect, which also dissipates more rapidly. Similarly, substituting the baseline shocks to the 10-year Treasury yield with alternative instruments yields results that closely align with those of the primary specification. Specifically, using shocks to the 30-year Treasury yield as an instrument produces nearly identical results while using shocks to the 5-year Treasury yield (weaker instrument) results in a somewhat more pronounced effect. These robustness checks generally affirm the validity and consistency of our baseline findings. However, they also highlight the need for caution in interpretation. While the main results largely hold in terms of direction and magnitude, it remains difficult to fully disentangle the direct effects of interest rate changes from the indirect influence of other macroeconomic factors, such as wage growth or stock market performance.

As an additional robustness check, we provide results using bootstrapped confidence intervals. This exercise is motivated by concerns about the validity of the LP-IV methodology, and more generally of instrumental variable regressions, as raised by Young (2022). In Figure A5 in the Appendix, the blue line indicates the original point estimate, while the green line shows the estimate obtained from the simulated data. The shaded areas represent bootstrapped confidence intervals constructed using the bootstrap-t method, following Young (2022). The dotted red lines correspond to the confidence intervals implied by the 2SLS residual bootstrap method. We use the same baseline specification as in our main analysis. Overall, we observe that the point estimate from the simulated data is generally indistinguishable from the original estimate. However, the bootstrapped confidence intervals are relatively wide, resembling the Anderson-Rubin confidence intervals. Nevertheless, the results obtained from the residual bootstrap are largely consistent with those from the more standard bootstrap-t method.

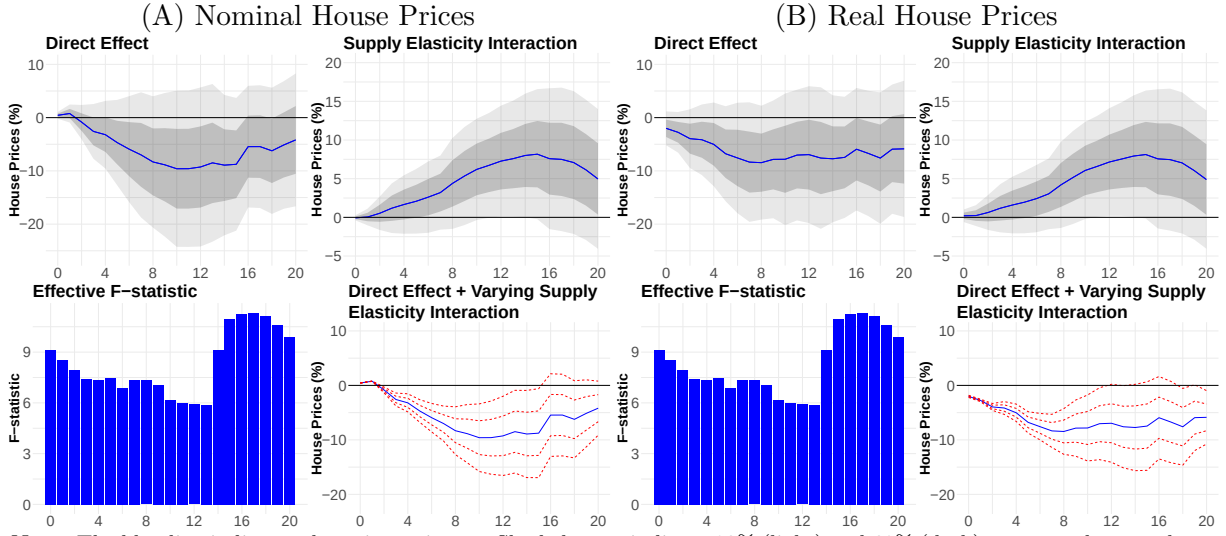
The results reveal a high degree of uncertainty, which is likely connected to potential nonlin-

³⁷Also, VAR models with uninformative priors are generally unstable with many variables.

earities and the influence of regional factors in house price formation. To address this, we now explore these aspects in greater depth. Figure 5 illustrates the panel regression results, incorporating the housing supply effect using the inverse housing supply elasticity from Guren *et al.* (2021). The figure presents the Impulse Response Functions (IRFs) of house prices in response to a one-percentage point increase in mortgage rates. Separate IRFs are shown for the direct effect and the supply elasticity interaction (indirect) effect. The blue line represents the point estimates, while the 90% confidence intervals (light grey bands) and 68% two-way cluster-robust confidence intervals (dark grey bands) provide measures of statistical uncertainty. Additionally, the effective F-statistic, calculated using a two-way cluster-robust covariance matrix as per Olea & Pflueger (2013), is displayed to assess instrument strength. The results are divided into two panels: the left panel shows the IRFs for nominal house prices, while the right panel depicts the IRFs for real house prices. For robustness, Figure A6 in the Appendix provides comparable results using the alternative housing supply instrument from Saiz (2010).

The top left panel of Figure 5 shows the impulse response for the direct effect, which should be roughly comparable to the results shown in the previous part, i.e., the IRF of house prices with the average supply elasticity. Similar to nationwide analysis, the maximal effect occurs between the third and fourth years, with the maximal semi-elasticity being approximately ten for both nominal and real house prices. The top right panel explores how the indirect effect varies with different levels of housing supply elasticity (expressed relative to its mean). We observe that the impact of housing supply elasticity is relatively small in the first year. However, over the longer run, housing supply has both a statically and economically significant effect on house prices, with the maximal indirect effect being close to ten. To better understand the combined direct and indirect effects, the bottom right panel (Direct Effect + Varying Supply Elasticity Interaction) presents scenarios with varying levels of housing supply elasticity (± 1 and ± 0.5 standard deviations). As intuition suggests, regions with constrained housing supply experience a more pronounced price drop, nearly twice as large. Figure A6 in the Appendix confirms our results as it shows that using an alternative housing supply instrument by Saiz (2010) provides almost identical results. Furthermore, Figures A7 and A8 in the Appendix present additional results using the FHFA dataset, which once again validate the robustness of our conclusions. Overall, these findings are broadly consistent with the asymmetric effects reported by Aastveit & Anundsen (2022), confirming the significance of housing supply constraints in moderating the price response to mortgage rate shocks. By contrast, Louie *et al.* (2025) offer a somewhat opposing perspective, arguing that higher income growth predicts similar house price growth across cities regardless of their estimated housing supply elasticity—i.e., the interaction term with income is not significant in their framework.

Figure 5: Impulse Response Functions (IRFs) and F-statistics for House Prices Using MSA Panel Data (Freddie Mac Dataset) with Guren Supply Elasticity Instrument

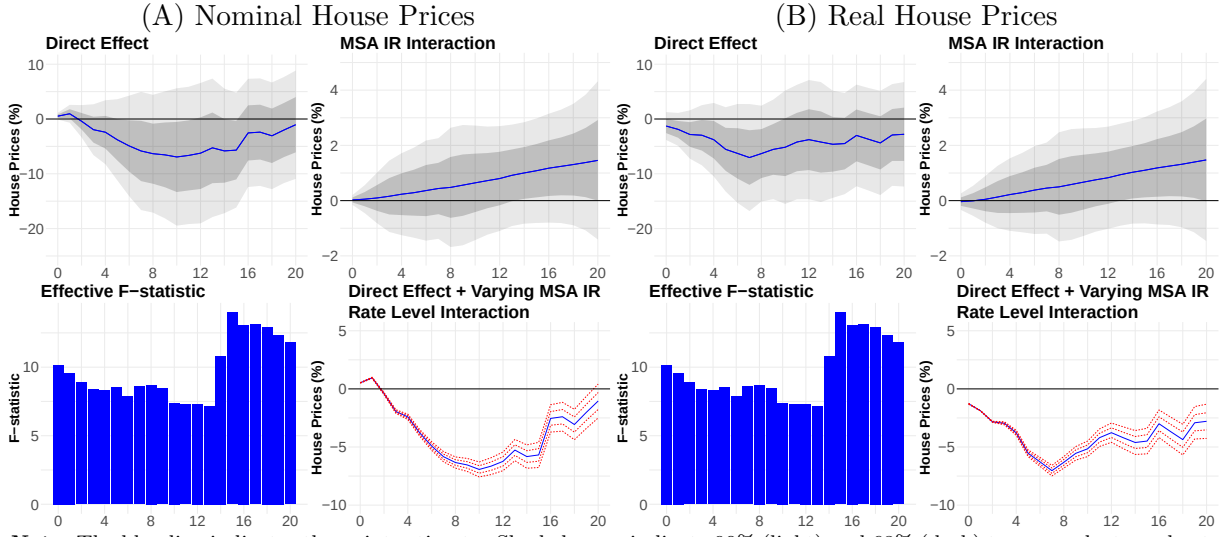


Note: The blue line indicates the point estimate. Shaded areas indicate 90% (light) and 68% (dark) two-way cluster-robust confidence intervals. The dotted lines show a range of scenarios with different supply elasticities (plus or minus 1 and 0.5 standard deviations). Effective F-statistic: First Stage Effective F-statistic is for the mortgage rate, and it is computed as in Olea & Pflueger (2013). 10 lags were used. The sample is 1993:Q1-2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

To address other potential sources of nonlinearities, we now examine the influence of regional interest rate levels. Figure 6 illustrates panel regression results that incorporate the effect of regional interest rate levels using Freddie Mac micro-data. Similar to the previous case, the figure displays Impulse Response Functions (IRFs) of house prices in response to a one-percentage point increase in mortgage rates. Separate IRFs are shown for the direct effect and the (MSA IR) interaction (indirect) effect of regional interest rate levels. We also display the combined direct and indirect effects (Direct Effect + Varying MSA IR Rate Level Interaction), which illustrate scenarios with varying regional levels of interest rates (± 1 and ± 0.5 standard deviations). This setup allows us to explore nonlinearities in house price responses to mortgage rate shocks, consistent with predictions from theoretical models. The results indicate that areas with higher average interest rates are less responsive to interest rate changes, consistent with theoretical predictions. The direct effect is slightly lower than in the previous case but remains close to 10, while the indirect effect (from a one-standard-deviation increase) is approximately 1.5—substantially larger than theoretical expectations but directionally aligned.³⁸ This suggests that our regional instruments may capture other influences. Additionally, Figure A9 in the Appendix presents robustness checks using the FHFA dataset, further confirming the robustness of these findings.

³⁸Both the overall standard deviation and the within-period standard deviations are approximately 0.1 percentage points.

Figure 6: Impulse Response Functions (IRFs) and F-statistics for House Prices Using MSA Panel Data (Freddie Mac Dataset) with Regional Interest Rate Level



Note: The blue line indicates the point estimate. Shaded areas indicate 90% (light) and 68% (dark) two-way cluster-robust confidence intervals. The dotted lines show a range of scenarios with different regional interest rate levels (plus or minus 1 and 0.5 standard deviations). Effective F-statistic: First Stage Effective F-statistic is for the mortgage rate, and it is computed as in Olea & Pflueger (2013). 10 lags were used. The sample is 1999:Q1-2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

7 Results - Housing Activity

We now move to study the effect of mortgage rates on other variables related to general housing activity (construction, permits, houses sold, houses for sale). As noted earlier, we use the same methodology (LP-IV, VAR, FVR³⁹) as with nationwide house prices.

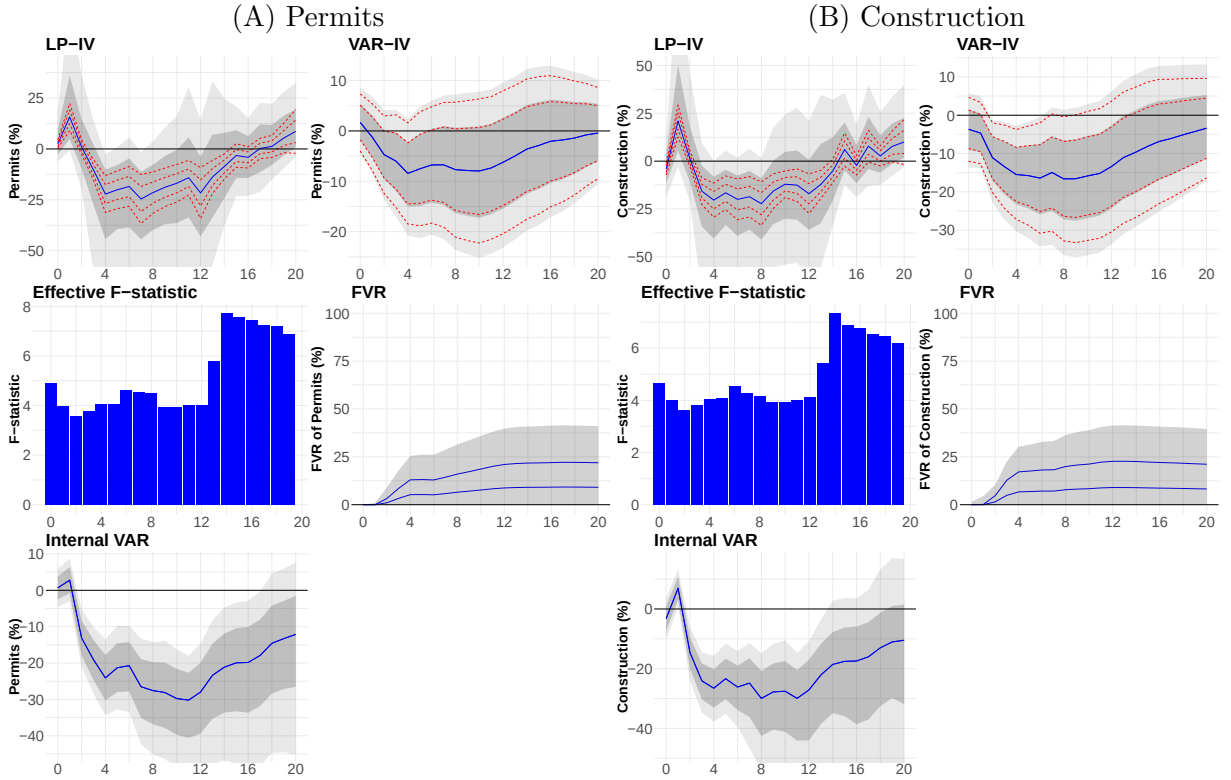
Figure 7 (left panel) analyzes housing permits, while the right panel focuses on construction starts. The results indicate that housing permits are significantly more sensitive to mortgage rate changes than house prices, exhibiting volatility approximately two to three times greater. Specifically, the LP-IV model estimates a maximal semi-elasticity of around 20, while the internal VAR model suggests a higher estimate closer to 30. The VAR-IV model provides a lower estimate, around 10, which may, however, be substantially biased. The use of these robust intervals is again appropriate given that the effective F-statistics is smaller than 10 for all periods. Therefore, estimates are accompanied by considerable uncertainty, as reflected in the 90% and 68% weak-instrument robust confidence intervals. The FVR indicates that mortgage rates contribute 10–25% to housing permits, though confidence intervals remain wide, with an upper bound of 40% and a lower bound of zero.

Similarly, results for construction activity (construction started) closely align with those for housing permits, indicating that housing starts exhibit a similar pattern of volatility, being two to three times more sensitive to mortgage rate changes than house prices. The estimated maximal semi-elasticity for housing starts is slightly lower than that for housing permits, generally

³⁹The pre-invertibility and bounds on the degree of invertibility R_0^2 imply non-invertibility in all cases as shown in the table A4 in the Appendix.

ranging from 20 to 30, depending on the model. (VAR-IV again indicates slightly lower semi-elasticity around 15, but may be biased.) However, as with housing permits, these estimates are characterized by wide confidence and credible intervals, reflecting a significant degree of uncertainty, which further underscores the statistical uncertainty associated with these results. The FVR indicates that the contribution of the mortgage rate to construction activity ranges from 10 to 25%, but with wide confidence intervals (the upper bound of confidence intervals is approximately 40%, and the lower bound is zero).

Figure 7: Impulse Response Functions (IRFs), Forecast Variance Ratio (FVR), and F-statistics for Construction and Permits



Note: LP-IV: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals constructed by inverting Anderson-Rubin (AR) tests. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the 2SLS estimator with HC standard errors. Effective F-statistic: First stage Effective F-statistic is computed as is Olea & Pflueger (2013) to assess instrument strength. Internal VAR: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) credible intervals. VAR-IV: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the delta-method. FVR: Forecast variance ratio, The blue lines indicate interval-identified estimates. Shaded areas indicate 90% confidence intervals. 8 lags were used in all models. The sample is 1988:Q2-2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

Figure 8 illustrates the impact of mortgage rates on the number of houses for sale and houses sold. For both variables, the effective F-statistics are generally below 8 throughout all periods, supporting the use of weak-instrument robust confidence intervals. The left panel focuses on houses for sale and reveals substantial variation across different models. The LP-IV and VAR models suggest that an increase in mortgage rates initially causes a rise in the number of houses for sale, with this trend persisting until the second or third year before declining. In contrast, the

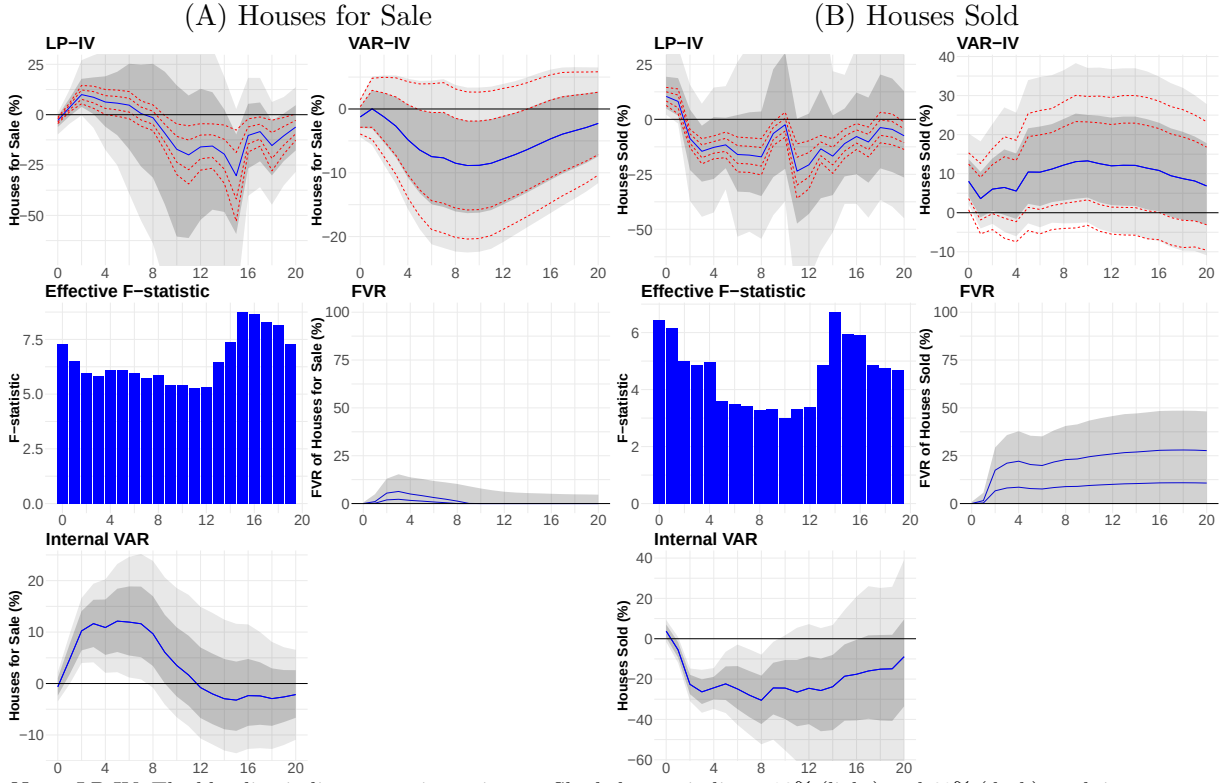
internal VAR model indicates a sharper initial increase, followed by a more pronounced decline. Meanwhile, the VAR-IV model displays an immediate decrease in the number of houses for sale after a mortgage rate hike. The forecast variance ratio (FVR) analysis suggests that the contribution of mortgage rates to the variation in the number of houses for sale is negligible. This divergence in results across methodologies highlights significant heterogeneity in the response of houses for sale during the analyzed period, potentially reflecting unique market dynamics such as the 'mortgage rate lock-in' effect observed in recent years.

The right panel of Figure 8 shows the effects of mortgage rates on houses sold. With the exception of the VAR-IV model, all methods consistently indicate that an increase in mortgage rates results in a decrease in the number of houses sold. However, the magnitude of this decline varies across models. The LP-IV model estimates a maximum decline of approximately 25%, while the internal VAR model suggests a slightly larger decrease of nearly 30%. Conversely, the VAR-IV model indicates an increase, though we caution readers that this estimate may be substantially biased, as previously noted. Across all models, the maximum decline in houses sold occurs between the second and fourth years following a mortgage rate hike. The FVR analysis shows that mortgage rates contribute to housing permits by approximately 10% to 25%, though these estimates are characterized by wide confidence intervals, underscoring significant uncertainty about the precise impact of mortgage rates on houses sold.

Broadly, excluding houses for sale, other housing activity variables—such as houses sold, construction activity, and permits—exhibit a sensitivity to mortgage rates that is two to three times greater than that observed for house prices. These findings are robust across different model specifications, as confirmed by our robustness checks (Figure A10). These checks include variations in lag structures and the addition of house prices as an additional covariate, alongside the lagged values of the dependent variable, unemployment, and inflation. These results reinforce the significant and amplified impact of mortgage rates on broader housing market dynamics compared to house prices alone.

Furthermore, as an additional robustness check, we provide results using bootstrapped confidence intervals. Figure A11 in the Appendix shows both the original point estimate and the estimate obtained from the simulated data. In addition, we report bootstrapped confidence intervals constructed using both the bootstrap-t method and the 2SLS residual bootstrap method. We again rely on the same baseline specification as in our main analysis of these variables. Overall, we find that the point estimate from the simulated data is essentially identical to the original estimate. The bootstrapped confidence intervals again resemble Anderson-Rubin confidence intervals, indicating relatively wide but robust uncertainty bands around the estimates.

Figure 8: Impulse Response Functions (IRFs), Forecast Variance Ratio (FVR), and F-statistics for Houses for Sale and Houses Sold



Note: LP-IV: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals constructed by inverting Anderson-Rubin (AR) tests. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the 2SLS estimator with HC standard errors. Effective F-statistic: First stage Effective F-statistic is computed as in Olea & Pflueger (2013) to assess instrument strength. Internal VAR: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) credible intervals. VAR-IV: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the delta-method. FVR: Forecast variance ratio, the blue lines indicate interval-identified estimates. Shaded areas indicate 90% confidence intervals. 10 lags were used in all models for Houses Sold. 8 lags were used in all models for Houses for Sale. The sample is 1988:Q2-2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

8 Conclusion

In this study, we examined the influence of mortgage rates on housing activity in the US, with a specific focus on house prices. More specifically, our main contribution to the existing literature is using shocks in long-term yields combined with an instrumental variable approach to assess the effect of mortgage rates on housing over time. The use of high-frequency shocks significantly reduces endogeneity issues, allowing for causal effect estimates. By using external instruments combined with a local projection and VAR frameworks, we were able to study how unexpected changes in mortgage rates influence housing activity, i.e., by explicitly distinguishing between mortgage and monetary policy rates, we enhanced clarity in understanding housing dynamics and the influence of mortgage rates.

Our results indicate that mortgage rates exert both statistically significant effects when considering 68% (weak-instrument robust) confidence/credible intervals and economically substantial impacts on house prices across most specifications and models. More specifically, the

maximal house price semi-elasticity to mortgage rates is generally close to 10 in most cases.⁴⁰ In other words, the results suggest that a downward one-percentage point shock to mortgage rates could lead to a cumulative 10% increase in house prices over the next three years, and conversely, an upward shock could result in a similar decline. Regarding the magnitude of the effect, it should also be mentioned that the short-run effect is relatively small, and it takes considerable time for the shock to propagate (two to three years) entirely. The delayed response of house prices may help explain why many previous studies report significantly smaller house price semi-elasticities in relation to changes in mortgage rates. On the other hand, our findings suggest that the effect of a mortgage rate increase is mainly transient in the long run (for most specifications), with most of the impact dissipating within a fifth-year horizon.

However, it is crucial to note that these estimates come with a high degree of uncertainty. For example, 90% weak-instrument robust confidence intervals generally do not show a statistically significant difference from zero. Given the considerable uncertainty, we estimate that the maximal implied semi-elasticity of house prices with respect to mortgage rates generally falls within a range of 5% to 15%. Additionally, the moving average model with external instruments (SVMA-IV) provides us with the set-identified variance decomposition (forecast variance ratios-FVRs) for the instrumented shock, which indicates mortgage rate shocks account for 2 to 25% of house price dynamics depending on the model specification.⁴¹ To put it plainly, our findings attribute up to 25% of the overall variation in house prices to decline in mortgage rates, or more generally to mortgage rate fluctuations. These decompositions underscore the significance of mortgage rate shocks in driving house price dynamics, albeit with considerable uncertainty, while also emphasizing the influence of long-term mortgage rates shaped by monetary policy, term premiums, and the natural rate of interest. Collectively, these factors have likely contributed to rising house prices, highlighting the need for policymakers to account for long-term rates when evaluating and addressing housing market dynamics.

Furthermore, we study the heterogeneity of these effects in different US regions, which allows us to partially differentiate the influences of housing supply, using supply elasticity results from Saiz (2010) and Guren *et al.* (2021). Our findings align with the results of Aastveit & Anundsen (2022), which highlight significant heterogeneity in house prices to monetary policy rates. In other words, suggesting that housing supply elasticity can significantly limit the effect of mortgage rates on house prices. Additionally, we analyze regional interest rate levels as another source of heterogeneity, finding that areas with higher average interest rates are less responsive to mortgage rate changes, consistent with theoretical predictions. Conversely, lower regional interest rate levels amplify the responsiveness of house prices. These results

⁴⁰Real house prices tend to respond somewhat more quickly to changes in mortgage rates, with the maximal elasticity being reached sooner. Moreover, the LP-IV and internal VAR models generally indicate higher semi-elasticities compared to the VAR-IV model. However, the Granger causality test (pre-invertibility test), as well as the continuous measure of the degree of invertibility R_0^2 in all specifications, implies that the instrumented mortgage rate is not fully invertible. This lack of invertibility suggests that the VAR-IV model may be substantially biased.

⁴¹For instance, using the FHFA index yields higher estimates (upper bound - 25%, lower bound - 10%). Case-Shiller index yields somewhat lower estimates (upper bound - 10%, lower bound - 2%). The results also vary slightly depending on whether real or nominal house price indices are used.

emphasize the importance of considering regional heterogeneity in interest rates when assessing the effects of monetary policy and mortgage rate changes on housing markets. Furthermore, our estimations confirm prior findings regarding house price semi-elasticity, showing that at average levels of housing supply elasticity or interest rates, the maximum house price semi-elasticity remains close to 10.

Finally, our analysis also demonstrates that mortgage rates have a significant influence on various dimensions of housing activity, including houses sold, construction, and building permits. Remarkably, with the exception of houses for sale, the sensitivity of these housing variables to changes in mortgage rates is generally two to three times greater than that of house prices. Specifically, across most specifications and models, unexpected changes in mortgage rates exhibit both statistically significant effects (based on weak-instrument robust 68% confidence intervals) and economically meaningful impacts on housing activity. The maximal effects of mortgage rates on these variables typically range from 10% to 30%, further highlighting their heightened volatility and responsiveness relative to house prices. Therefore, our results suggest that a downward one-percentage point shock to mortgage rates could lead to a cumulative 20 - 30% increase in housing construction, building permits, and number of houses sold over the next three years, and vice versa, an upward shock could result in a similar decrease. Furthermore, our findings also attribute up to 10% to 30% of the overall variation in houses sold, construction, and building permits to changes in mortgage rates. This underscores the critical role mortgage rates play in shaping housing market dynamics.

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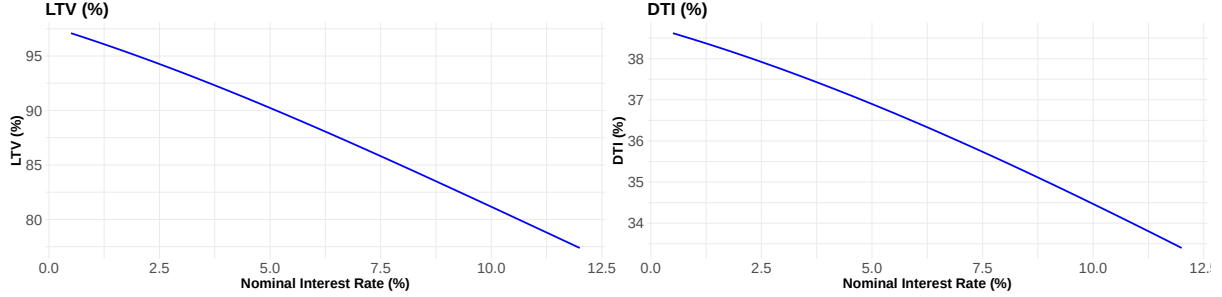
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A Appendix

Figure A1: Partial Equilibrium Model: LTV, DTI



Note: The left panel shows the size of the LTV as a function of interest rate. The right panel shows the size of the size of the mortgage payment to the income (DTI) as a function of interest rate. (Model calibration: $M = 30$, $\beta = 0.9883$, $\xi = 0.6$)

Table A1: Descriptive Statistics: House Prices (FHFA Dataset)

Variable		Mean	Sd.	Min	Max	N
(log) Real House Prices	Overall	-0.119	0.26	-0.83	1.01	10692
(log) Real House Prices	Between		0.15	-0.42	0.31	81
(log) Real House Prices	Within		0.22	-0.72	0.69	132
Δ (log) Nominal House Price	Overall	1.08	2.19	-15.07	11.84	10611
Δ (log) Nominal House Prices	Between		0.20	0.68	1.55	81
Δ (log) Nominal House Prices	Within		2.18	-14.95	11.89	131
Δ (log) Real House Prices	Overall	0.44	2.17	-16.93	10.88	10611
Δ (log) Real House Prices	Between		0.20	0.05	0.91	81
Δ (log) Real House Prices	Within		2.16	-16.81	10.94	131

Note: The table displays summary statistics regarding the house price data used in the panel estimation. Sd. denotes standard deviation. N denotes the number of observations. The table uses data regarding the FHFA purchase-only dataset for the 100 Largest Metropolitan Statistical Areas. The sample is 1993:Q1-2023:Q4.

Table A2: First Stage Regression Results

	Δ 30-year Mortgage rate							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
TNOTE10	2.101*** (0.538)	1.951*** (0.647)						
MPS			1.125** (0.468)	0.635 (0.429)				
TNOTE05					1.612*** (0.495)	1.331*** (0.503)		
TBOND							2.057*** (0.593)	2.074*** (0.757)
Constant	-0.010 (0.039)	-1.075 (1.709)	-0.020 (0.038)	-1.424 (1.767)	-0.006 (0.040)	-0.956 (1.727)	-0.023 (0.039)	-1.375 (1.694)
Controls	NO	YES	NO	YES	NO	YES	NO	YES
Observations	134	134	134	134	134	134	134	134
R ²	0.080	0.354	0.060	0.319	0.075	0.346	0.059	0.345
Adjusted R ²	0.073	0.150	0.053	0.103	0.068	0.138	0.051	0.138
Effective F Statistic	12.31	9.28	6.46	2.36	10.46	7.22	9.93	7.60

Note: HAC standard errors with 1 lag. *p<0.1; **p<0.05; ***p<0.01. TNOTE10 denotes shock to the 10-year Treasury note. TNOTE05 denotes shock to the 5-year Treasury note. TBOND denotes shocks to the 30-year Treasury bond. MPS denotes monetary policy surprises computed as the first principal component of shocks in short-term FED funds futures and two to four-quarter-ahead Eurodollar futures (Nakamura & Steinsson, 2018a). The controls include 10 lags of the nominal Case-Shiller index, unemployment rate, inflation, and linear trend. Effective F-statistic is computed as in Olea & Pflueger (2013) to assess instrument strength.

Table A3: First Stage Regression Results - Subsample Results

Δ 30-year Mortgage rate (1988 - 2010)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
TNOTE10	1.580*** (0.570)	1.597** (0.685)						
MPS			0.623 (0.484)	0.313 (0.423)				
TNOTE05					1.087* (0.626)	0.878 (0.585)		
TBOND							1.674*** (0.624)	2.007*** (0.726)
Constant	-0.055 (0.045)	-2.613 (1.978)	-0.058 (0.045)	-2.497 (2.012)	-0.052 (0.046)	-2.645 (1.965)	-0.068 (0.044)	-3.322* (1.913)
Controls	NO	YES	NO	YES	NO	YES	NO	YES
Observations	88	88	88	88	88	88	88	88
R ²	0.042	0.237	0.021	0.210	0.029	0.220	0.038	0.247
Adjusted R ²	0.031	0.090	0.009	0.058	0.018	0.071	0.027	0.102
Effective F Statistic	5.07	5.60	1.70	0.55	2.81	2.43	4.98	7.52
Δ 30-year Mortgage rate (2011 - 2023)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
TNOTE10	3.080** (1.202)	2.963** (1.447)						
MPS			2.644** (1.271)	2.138* (1.287)				
TNOTE05					2.369*** (0.863)	2.246** (1.062)		
TBOND							3.070** (1.463)	2.722 (1.776)
Constant	0.079 (0.071)	0.632 (11.016)	0.033 (0.059)	3.394 (9.940)	0.082 (0.072)	1.441 (10.482)	0.069 (0.070)	-1.951 (11.905)
Controls	NO	YES	NO	YES	NO	YES	NO	YES
Observations	48	48	48	48	48	48	48	48
R ²	0.176	0.503	0.245	0.483	0.186	0.505	0.118	0.452
Adjusted R ²	0.158	0.292	0.229	0.264	0.169	0.294	0.099	0.220
Effective F Statistic	6.81	4.39	4.69	2.78	7.97	4.77	4.71	2.46

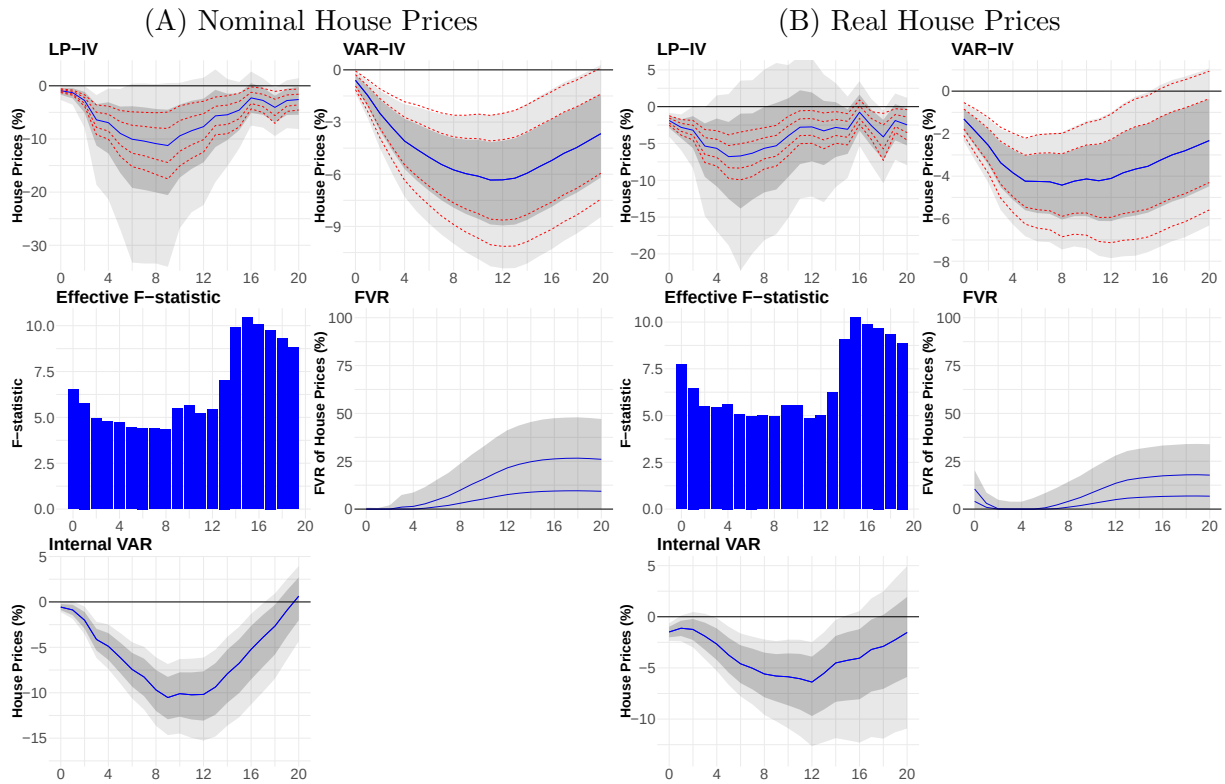
Note: HAC standard errors with 1 lag. *p<0.1; **p<0.05; ***p<0.01. TNOTE10 denotes shock to the 10-year Treasury note. TNOTE05 denotes shocks to the 5-year Treasury note. TBOND denotes shock to the 30-year Treasury bond. MPS denotes monetary policy surprises computed as the first principal component of shocks in short-term FED funds futures and two to four-quarter-ahead Eurodollar futures (Nakamura & Steinsson, 2018a). The controls include 4 lags of the nominal Case-Shiller index, unemployment rate, inflation, and linear trend. Effective F-statistic is computed as in Olea & Pflueger (2013) to access instrument strength.

Table A4: Degree of Invertibility/Recoverability

	R_0^2 (Invertibility)	R_∞^2 (Recoverability)	Pre-invertibility (Granger) test
Nominal House Prices (Case-Shiller Index)			
Bound estimates	0.09-0.241	0.367-1	
90% confidence interval	0-0.377	0.245-1	
p-value			0.09
Real House Prices (Case-Shiller Index)			
Bound estimates	0.083-0.224	0.371-1	
90% confidence interval	0-0.359	0.252-1	
p-value			0.01
Nominal House Prices (FHFA index)			
Bound estimates	0.074-0.199	0.377-1	
90% confidence interval	0-0.344	0.255-1	
p-value			0.001
Real House Prices (FHFA Index)			
Bound estimates	0.094-0.223	0.41-1	
90% confidence interval	0-0.371	0.284-1	
p-value			0.023
Permits			
Bound estimates	0.111-0.265	0.419-1	
90% confidence interval	0.01-0.413	0.312-1	
p-value			0
Construction (Started)			
Bound estimates	0.129-0.324	0.395-1	
90% confidence interval	0.016-0.488	0.287-1	
p-value			0
Houses for Sale			
Bound estimates	0.111-0.284	0.387-1	
90% confidence interval	0.01-0.428	0.276-1	
p-value			0.13
Houses Sold			
Bound estimates	0.098-0.248	0.39-1	
90% confidence interval	0-0.39	0.289-1	
p-value			0.003

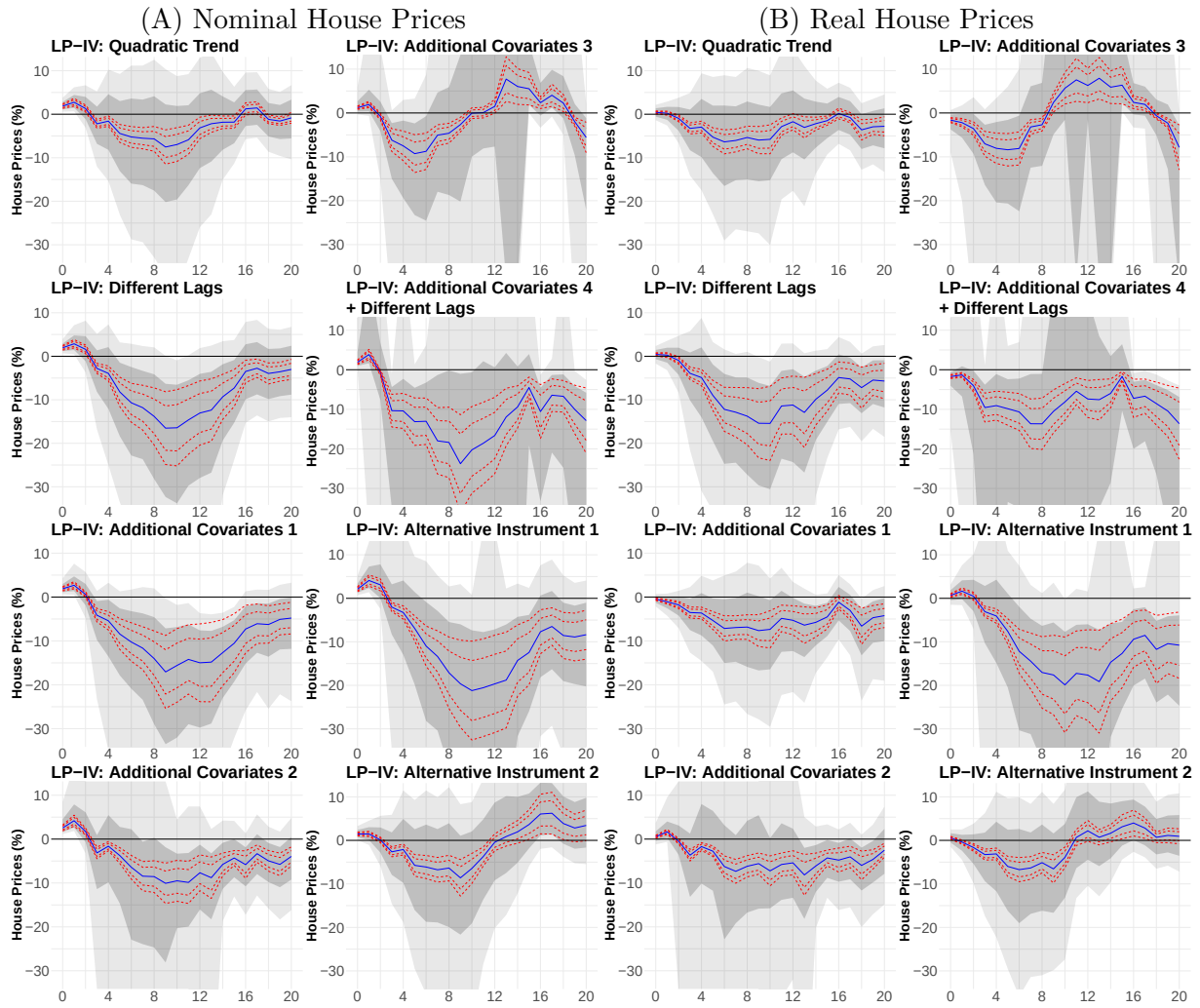
Note: Bounds on the degree of invertibility R_0^2 and the degree of recoverability R_∞^2 . Ten (House Prices, Houses Sold) or eight (House Permits, Construction, Houses for Sale) lags were used.

Figure A2: Robustness Check 1: Impulse Response Functions (IRFs), Forecast Variance Ratio (FVR), and F-statistics for House Prices (FHFA Index)



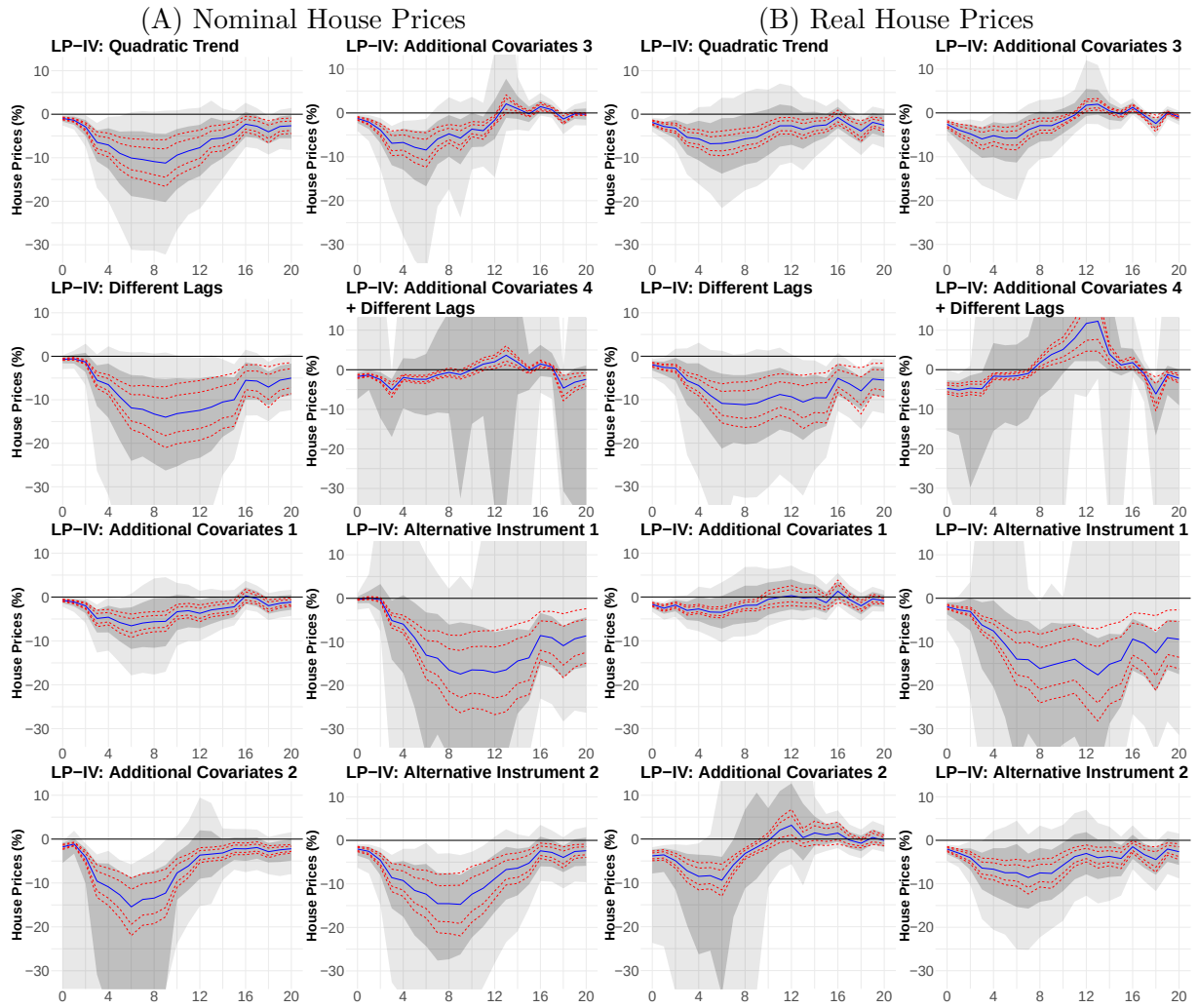
Note: LP-IV: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals constructed by inverting Anderson-Rubin (AR) tests. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the 2SLS estimator with HC standard errors. Effective F-statistic: First stage Effective F-statistic is computed as is Olea & Pflueger (2013) to assess instrument strength. Internal VAR: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) credible intervals. VAR-IV: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the delta-method. FVR: Forecast variance ratio, the blue lines indicate interval-identified estimates. Shaded areas indicate 90% confidence intervals. 10 lags were used in all models. The sample is 1991:Q1-2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

Figure A3: Robustness Check 2: Impulse Response Functions (IRFs) for House Prices (Case–Shiller Index)



Note: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals constructed by inverting Anderson-Rubin (AR) tests. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the 2SLS estimator with HC standard errors. LP-IV: Quadratic Trend: uses quadratic trend. LP-IV: Different Lags: uses an alternative lag structure with 8 lags. LP-IV: Additional Covariates 1: includes lagged dependent variable, inflation, unemployment, houses sold, and houses for sale. LP-IV: Additional Covariates 2: includes lagged dependent variable, inflation, unemployment, housing starts, and the number of permits. LP-IV: Additional Covariates 3: includes lagged dependent variable, inflation, unemployment, stock prices, and real wages. LP-IV: Additional Covariates 4: includes lagged dependent variable, inflation, unemployment, permits, stock prices, and real wages while also using 8 lags. LP-IV: Alternative Instrument 1: uses shocks to 5-year Treasury note as an instrument. LP-IV: Alternative Instrument 2: uses shocks to 30-year Treasury bond as an instrument. The sample is 1988:Q2–2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

Figure A4: Robustness Check 3: Impulse Response Functions (IRFs) for House Prices (FHFA Index)



Note: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals constructed by inverting Anderson-Rubin (AR) tests. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the 2SLS estimator with HC standard errors. LP-IV: Quadratic Trend: uses quadratic trend. LP-IV: Different Lags: uses an alternative lag structure with 8 lags. LP-IV: Additional Covariates 1: includes lagged dependent variable, inflation, unemployment, houses sold, and houses for sale. LP-IV: Additional Covariates 2: includes lagged dependent variable, inflation, unemployment, housing starts, and the number of permits. LP-IV: Additional Covariates 3: includes lagged dependent variable, inflation, unemployment, stock prices, and real wages. LP-IV: Additional Covariates 4: includes lagged dependent variable, inflation, unemployment, permits, stock prices, and real wages while also using 8 lags. LP-IV: Alternative Instrument 1: uses shocks to 5-year Treasury note as an instrument. LP-IV: Alternative Instrument 2: uses shocks to 30-year Treasury bond as an instrument. The sample is 1991:Q1-2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

Figure A5: Robustness Check 4: Impulse Response Functions (IRFs) for House Prices (Bootstrapped Confidence Intervals)

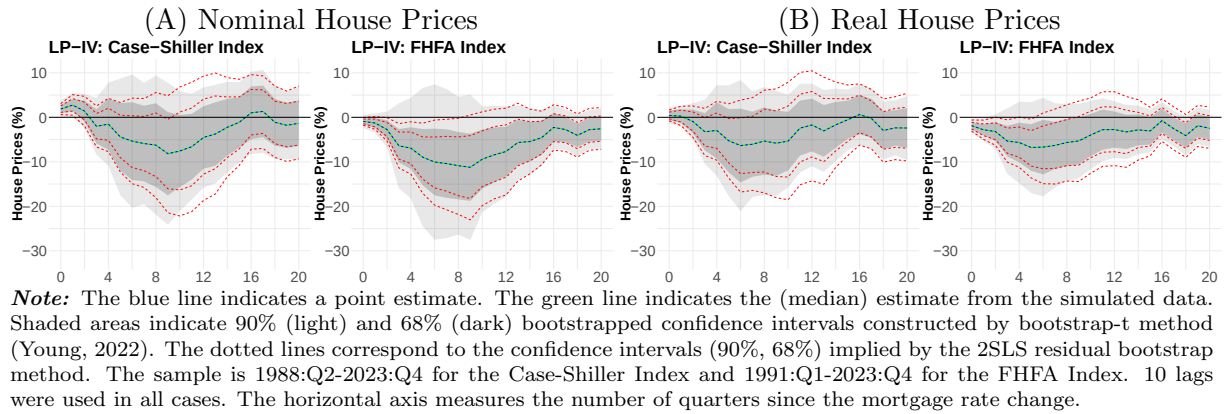


Figure A6: Robustness Check 5: Impulse Response Functions (IRFs) and F-statistics for House Prices Using MSA Panel Data (Freddie Mac Dataset) with Saiz Supply Elasticity Instrument

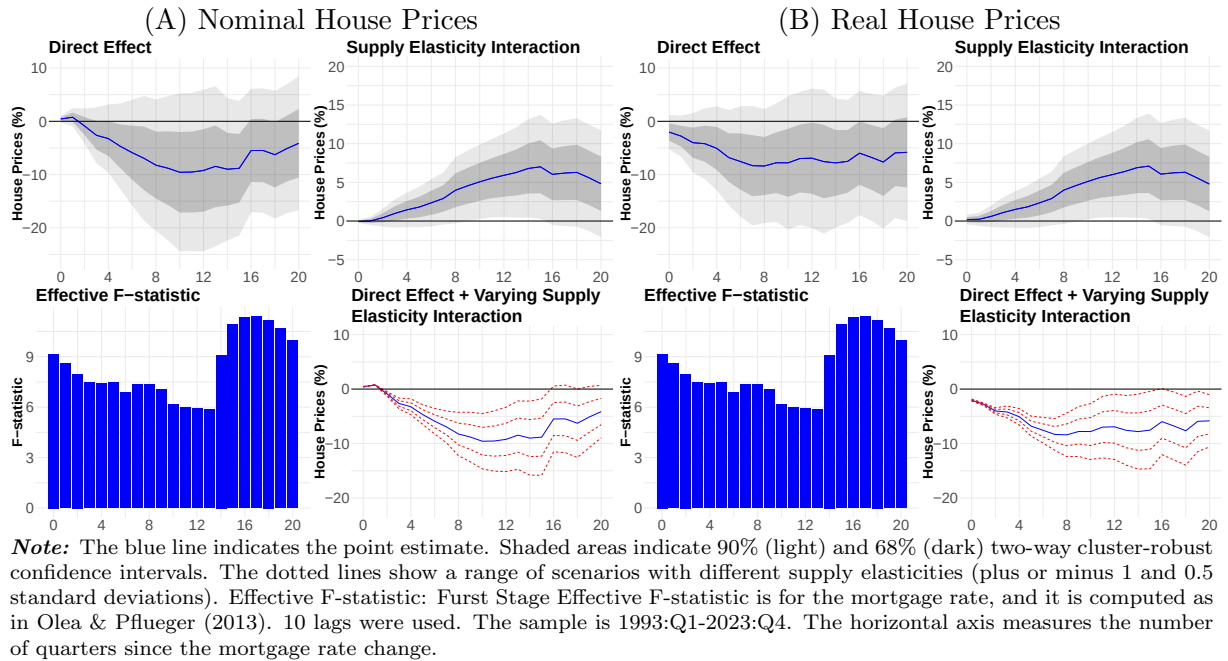
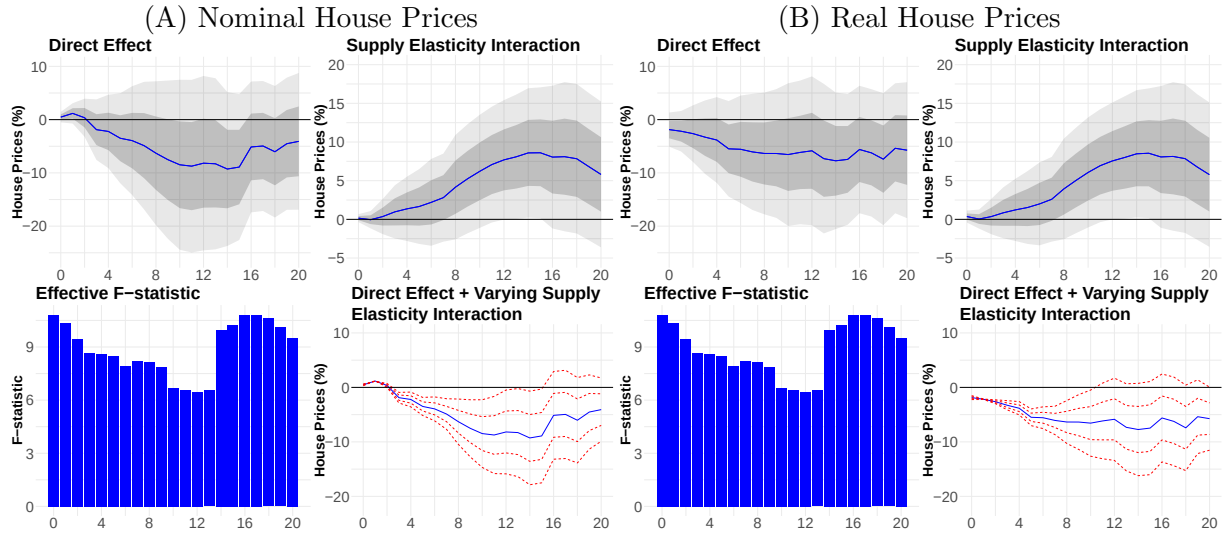
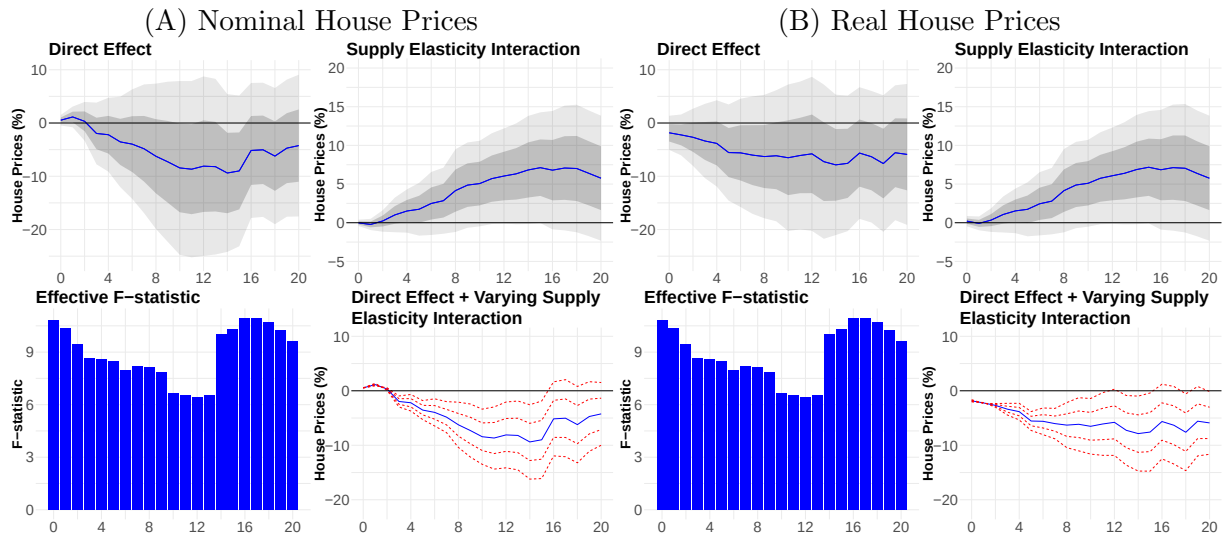


Figure A7: Robustness Check 6: Impulse Response Functions (IRFs) and F-statistics for House Prices Using MSA Panel Data (FHFA Dataset) with Guren Supply Elasticity Instrument



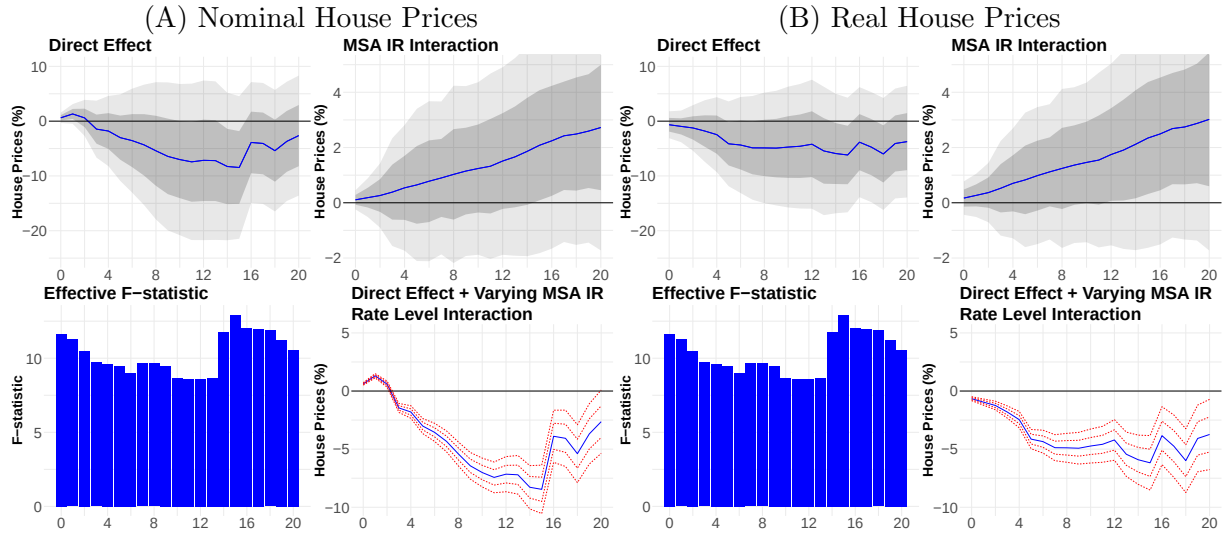
Note: The blue line indicates the point estimate. Shaded areas indicate 90% (light) and 68% (dark) two-way cluster-robust confidence intervals. The dotted lines show a range of scenarios with different supply elasticities (plus or minus 1 and 0.5 standard deviations). Effective F-statistic: First Stage Effective F-statistic is for the mortgage rate, and it is computed as in Olea & Pflueger (2013). 10 lags were used. The sample is 1993:Q1-2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

Figure A8: Robustness Check 7: Impulse Response Functions (IRFs) and F-statistics for House Prices Using MSA Panel Data (FHFA Dataset) with Saiz Supply Elasticity Instrument



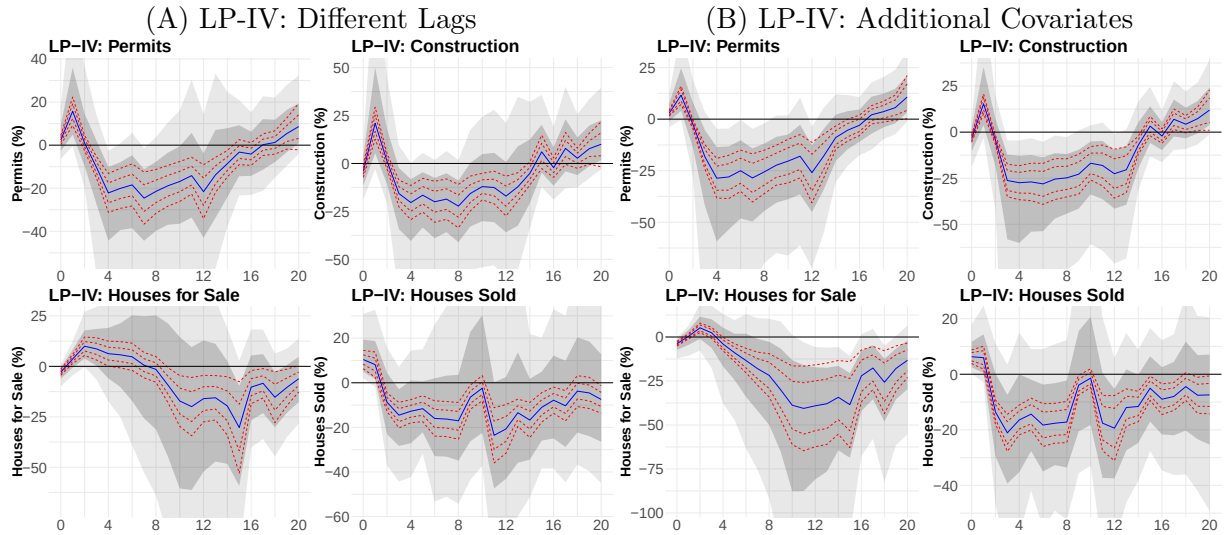
Note: The blue line indicates the point estimate. Shaded areas indicate 90% (light) and 68% (dark) two-way cluster-robust confidence intervals. The dotted lines show a range of scenarios with different supply elasticities (plus or minus 1 and 0.5 standard deviations). Effective F-statistic: First Stage Effective F-statistic is for the mortgage rate, and it is computed as in Olea & Pflueger (2013). 10 lags were used. The sample is 1993:Q1-2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

Figure A9: Robustness Check 8: Impulse Response Functions (IRFs) and F-statistics for House Prices Using MSA Panel Data (FHFA dataset) with Regional Interest Rate Level



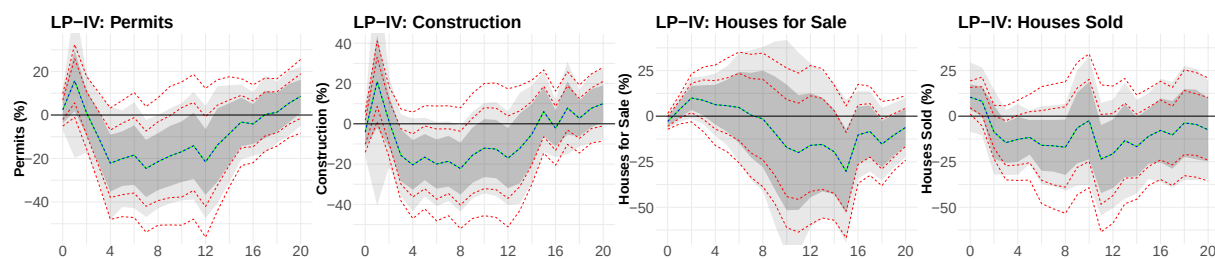
Note: The blue line indicates the point estimate. Shaded areas indicate 90% (light) and 68% (dark) two-way cluster-robust confidence intervals. The dotted lines show a range of scenarios with different regional interest rate levels (plus or minus 1 and 0.5 standard deviations). Effective F-statistic: First Stage Effective F-statistic is for the mortgage rate, and it is computed as in Olea & Pflueger (2013). 10 lags were used. The sample is 1999:Q1-2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

Figure A10: Robustness Check 9: Impulse Response Functions (IRFs) for Housing Activity Variables



Note: The blue line indicates a point estimate. Shaded areas indicate 90% (light) and 68% (dark) weak-instrument robust confidence intervals constructed by inverting Anderson-Rubin (AR) tests. The dotted lines correspond to the confidence intervals (90%, 68%) implied by the 2SLS estimator with HC standard errors. LP-IV: Different Lags: uses an alternative lag structure with 8 (Houses Sold) or 10 (Permits, Construction, Houses for Sale) lags. LP-IV: Additional Covariates: includes lagged dependent variable, inflation, unemployment, and nominal house prices (Case-Shiller index). The sample is 1991:Q1-2023:Q4. The horizontal axis measures the number of quarters since the mortgage rate change.

Figure A11: Robustness Check 10: Impulse Response Functions (IRFs) for Housing Activity Variables (Bootstrapped Confidence Intervals)



Note: The blue line indicates a point estimate. The green line indicates the (median) estimate from the simulated data. Shaded areas indicate 90% (light) and 68% (dark) bootstrapped confidence intervals constructed by bootstrap-t method (Young, 2022). The dotted lines correspond to the confidence intervals (90%, 68%) implied by the 2SLS residual bootstrap method. The sample is 1988:Q2-2023:Q4. 8 lags were used except for Houses Sold (10 lags). The horizontal axis measures the number of quarters since the mortgage rate change.